

# CONTENTS

## SIGNIFICANT SMALL SHIPS OF 2015

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### INTRODUCTION

|                                                                                              |    |
|----------------------------------------------------------------------------------------------|----|
| <b>ALUSAFE 1100 FIC:</b> Speed freak of a patrol craft makes debut in prototype form         | 2  |
| <b>ALYA MCCALL:</b> Offshore crew/supply transfer vessel concept goes large                  | 4  |
| <b>ALYSEE:</b> Customised mussel farming craft                                               | 6  |
| <b>AUGUST NAERØ:</b> Low-noise, first entry in Dockstavarvet's IC 20 M class                 | 8  |
| <b>BENREOCH:</b> CTV designed for smooth interactions with offshore platforms                | 12 |
| <b>CLEAN HARBOUR 1 &amp; 2:</b> Hong Kong's first domestic diesel-electric container vessels | 14 |
| <b>CURTIS MARSHALL:</b> Compact dive support vessel, optimised for offshore wind work        | 16 |
| <b>ERNST MEIER-HEDE:</b> SAR and fire-fighting craft                                         | 20 |
| <b>GOBBLER 290:</b> Revolutionary tank-free oil skimmer prototype                            | 22 |
| <b>GUSTAF AF KLINT:</b> Unique 'clear sweeping' vessel                                       | 24 |
| <b>HELIOTROPE 48:</b> Pleasure cat built to blend high comfort with high performance         | 26 |
| <b>ICENI VENTURE:</b> First South Boats IOW South Cat 23 crew transfer vessel                | 28 |
| <b>IDAR ULSTEIN:</b> Latest high-tech addition to Norwegian Sea Rescue's fleet               | 30 |
| <b>KAROLINE:</b> First battery-powered fishing boat                                          | 32 |
| <b>L'ESCAPADE:</b> French hybrid power passenger boat                                        | 36 |
| <b>LONDON TITAN:</b> Multipurpose boat for maintenance, support and dredging work            | 38 |
| <b>MAPLIN:</b> First hydrographic survey vessel from CTruk                                   | 42 |
| <b>NEIL ARMSTRONG:</b> New oceanographic survey vessel                                       | 44 |
| <b>OCEAN EAGLE 43:</b> Trimaran offshore patrol vessel with monohull-style motions           | 46 |
| <b>ORMSÖ:</b> Ice-breaking double-ended car and passenger ferry                              | 48 |
| <b>OSSOY:</b> first medium-sized AHTS class constructed by Sinopacific                       | 50 |
| <b>PELICAN PRIDE:</b> Economic offshore FSIV with additional creature comforts               | 52 |
| <b>PRINCE JOB I:</b> New multipurpose OSV for African offshore fields                        | 54 |
| <b>RT EMOTION:</b> First hybrid tug to operate within German waters                          | 56 |
| <b>SISTAC VITORIA:</b> Monohull dive support vessel hits Brazilian offshore waters           | 58 |
| <b>TAI PARI:</b> First of two customised China-built tugs with optimised Z-drives            | 60 |
| <b>TREARDUR BAY:</b> First craft to install VLJ propulsion technology                        | 62 |
| <b>TRIDENT:</b> Versatile research vessel for academic projects and training                 | 64 |
| <b>UMOE VENTUS:</b> Wind farm support craft combining hovercraft and cat technology          | 66 |
| <b>YENİÇAY 1:</b> 'Big sister' to RAL's previous RAscal 1500 class                           | 68 |
|                                                                                              | 70 |



M/V Alya McCall | 206', 100 Passenger, DP-2 Monohull Fast Support Vessel Builder: Gulf Craft LLC Operator: SEACOR Marine Speed @ 50t deadweight: 38 knots



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## ALUSAFE 1100 FIC: Speed freak of a patrol craft makes debut in prototype form

|                                                      |                         |
|------------------------------------------------------|-------------------------|
| Builder .....                                        | <b>Maritime Partner</b> |
| Designer .....                                       | <b>Maritime Partner</b> |
| Vessel's name .....                                  | <b>Alusafe 1100 FIC</b> |
| Owner/operator .....                                 | <b>Maritime Partner</b> |
| Country .....                                        | <b>Norway</b>           |
| Flag .....                                           | <b>Norway</b>           |
| Total number of sister ships already completed ..... | <b>0</b>                |
| Total number of sister ships still on order .....    | <b>0</b>                |
| Contract date .....                                  | <b>N/A</b>              |
| Delivery date .....                                  | <b>N/A</b>              |

**M**aking its debut at the Seawork International Exhibition in June 2015, Alusafe's 1100 fast interceptor craft (FIC) prototype certainly captured visitor attention – due, in part, to its striking livery, and the high speeds offered by this new class of patrol boat.

Designed and constructed by Norway's Maritime Partner, and intended to be pushed to the limit by coast guard / police agencies and naval fleet operators alike, the Alusafe 1100 FIC features a planing monohull with a single chine and deep-vee hullform, intended to assist the crew in cutting through inclement weather at a service speed of 50knots, increasing to at least 60knots max. The hull has been fashioned from marine grade aluminium and is reinforced by longitudinal stiffeners, as well as transverse frames and bulkheads.

At this stage of development, pre-order, the vessel relies on a pair of Mercury Verado 400 outboards; however, Maritime Partner states, the craft can also be configured to operate with three outboards, should higher speeds and extra kick be required.

One of the Alusafe 1100 FIC's key design features has been the 'puncture-proof' nature of its modular fender system, which comprises fender modules, mounting hardware and fixing profiles. The boat can be fitted with ballistic protection to various levels, and a host of different rapid entry/boarding systems can be integrated into the class. Maritime Partner has also ensured that the vessel will not be ducking out of action prematurely; the inclusion of two petrol tanks, each with a capacity of 550litres, has enabled the vessel to attain a range of 225nm.

The Alusafe 1100 FIC is just one in a range of new FIC proposed by Maritime Partner; other models include the Alusafe 1200 FIC, which adds the presence of a cabin, and the Alusafe 1400 FIC, designed for operations in sea state 4 and capable of accommodating four crew members and up to 12 passengers/casualties.

### TECHNICAL PARTICULARS

|                  |       |
|------------------|-------|
| Length, oa ..... | 11.3m |
| Length, bp ..... | 11.2m |

|                                    |                        |
|------------------------------------|------------------------|
| Breadth, moulded .....             | 2.82m                  |
| Depth, moulded .....               | 0.95m (with engines)   |
| Gross tonnage .....                | 3.5tonnes              |
| Service speed .....                | 50knots (80% MCR)      |
| Max speed .....                    | 60knots                |
| Range (nautical miles) .....       | 225                    |
| Main engine(s)                     |                        |
| Make .....                         | Mercury                |
| Model .....                        | Verado 400             |
| Number .....                       | 2                      |
| Output of each engine .....        | 298kW                  |
| Propeller(s)                       |                        |
| Material .....                     | Stainless steel        |
| Manufacturer .....                 | Mercury Marine Bravo 1 |
| Number .....                       | 1                      |
| Fixed/controllable pitch .....     | Fixed                  |
| Open or nozzleed .....             | Open                   |
| Bridge electronics                 |                        |
| Radar(s) .....                     | Simrad                 |
| GMDSS .....                        | Simrad                 |
| Other communications systems ..... | VHF Simrad 6210        |
| GPS .....                          | Simrad                 |
| Chart plotter .....                | Simrad                 |
| Onboard capacities                 |                        |
| Fuel oil .....                     | 1,100litres            |
| Complement                         |                        |
| Crew .....                         | 2                      |
| Passengers .....                   | 6                      |
| Number of cabins .....             | 0                      |





## ALYA McCALL: Offshore crew/supply transfer vessel concept goes large

Builder ..... **Gulf Craft**  
 Designer ..... **Incat Crowther**  
 Vessel's name ..... **Alya McCall**  
 Owner/operator ..... **Seacor Marine**  
 Country ..... **US**  
 Flag ..... **US**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **1**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **October 2015**

As the first of two new crew and supply transfer vessels delivered to offshore operator Seacor Marine, for deployment in Saudi Arabia, the 62.8m loa *Alya McCall* captured the marine sector's attention in Q4 2015, raising the bar for these vessel types by dint of her size.

Far larger than standard CTVs or supply craft, *Alya McCall* represents the debutant in the Incat Crowther-designed and Gulf Craft-built Express Plus class. The vessel was planned and pieced together in Louisiana, US, home to both Gulf Craft's shipbuilding facilities and Incat Crowther's US office.

*Alya McCall*, measures 62.79m loa (58.9m wl), 10.06m in breadth and 4.6m in depth, and has been designed to carry up to 16 crew members and 100 passengers. Fashioned from marine grade aluminium, the US-flagged vessel draws 2.8m. Onboard capacities include 316,000litres of fuel oil and 28,000litres of fresh water, and the main deck covers an area of 328m<sup>2</sup>, strengthened to house approximately 300tonnes' worth of cargo.

The vessel is powered by five Cummins QSK60-M main engines, rated 1,998kW apiece,

driving five Hamilton HT-810 waterjets through Twin Disc MGX 61500 SC reverse reduction gearboxes. The gearboxes are connected to the jets via a cardan shafting system provided by Driveline Service. Of the HT-810 units, the jet positioned on the boat's centre line will not feature steering or reverse buckets, as its primary function is to beef up the vessel's speed.

In sea trials, the above configuration was shown to grant *Alya McCall* a top speed of 38knots in lightship mode. When fully loaded, the vessel's speed would be more in the region of 20knots, although, by Seacor Marine's admission, those seeking high / lightship speeds are the target market for the Express Plus range.

The vessel has also been fitted with three Cummins QSM 11 generators (each rated 290kW) for onboard power, and a trio of 149kW bow thrusters, produced by Thrustmaster, for enhanced station-keeping capability. The thrusters and jets are controlled by a dynamic positioning 2 (DP2) system, supplied by Kongsberg. Seacor Marine claims that this thruster arrangement provides an added layer of redundancy; should one bow thruster fail, the vessel will be able to maintain its DP2 status. For crew and passenger comfort, Gulf Craft has also installed a Naiad Dynamics ride control system aboard the vessel, while FFS provided her fire pump system.

### TECHNICAL PARTICULARS

Length, oa ..... 62.8m  
 Length, bp ..... 58.9m  
 Breadth, moulded ..... 9.8m  
 Depth, moulded ..... 4.6m  
 Gross tonnage ..... 98tonnes

Displacement ..... 700tonnes  
 Design, draught ..... 2.8m  
 Design, deadweight ..... 350tonnes  
 Lightweight ..... 350tonnes  
 Deck space (total) ..... 328m<sup>2</sup>  
 Deck capacity ..... 2.64tonnes/m<sup>2</sup>  
 Service speed ..... 33knots @ 150tonnes  
 Max speed ..... 38knots  
 Classification society ..... ABS  
 Notations ..... ABS ✱A1 HSC Crewboat  
 ✱AMS ✱DP2 Fire Fighting Capable  
 USCG Subchapter T, Oceans

Main engine(s)  
 Make ..... Cummins  
 Model ..... QSK-60  
 Number ..... 5  
 Output of each engine ..... 1,998kW @ 1,900rpm  
 Gearbox(es)  
 Make ..... Twin Disc  
 Model ..... MGX 61500 SC  
 Number ..... 5  
 Waterjet(s)  
 Make ..... Hamilton HT-810  
 Number ..... 5  
 Bow thruster(s)  
 Make ..... Thrustmaster 30TT200ML  
 Number ..... 3  
 Deck machinery ..... 2 x FFS firefighting pumps  
 Onboard capacities  
 Fuel oil ..... 316,000litres  
 Fresh water ..... 28,000litres  
 Sullage ..... 4,540litres  
 Complement  
 Crew ..... 16  
 Passengers ..... 100  
 Number of cabins ..... 6



## ALYSÉE: Stability and manoeuvrability are key for new customised mussel farming craft

Builder ..... **Alnmaritec Ltd**  
 Designer ..... **Alnmaritec Ltd**  
 Vessel's name ..... **Alysée**  
 Owner/operator ..... **Offshore Shellfish Ltd**  
 Country ..... **UK**  
 Flag ..... **UK**  
 Total number of sister ships  
     already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **September 2014**  
 Delivery date ..... **May 2015**

*Alysée* represents the first (and possibly a one-off) in Alnmaritec's new 15.2m aluminium Wave Farmer 1400 mussel farming boat class, which the UK yard designed and assembled for operator Offshore Shellfish. This series has been described as a development of the builder's established Wave Train catamaran hull design, which includes fine forward sections developing into a medium deadrise, single chine hull form. However, given Offshore Shellfish's specific operational requirements off the UK's south-western coast, *Alysée* has been customised precisely to meet the tasks at hand.

For example, one major requirement with this new vessel was that she would be able to access Offshore Shellfish's rope-grown mussel farms, situated in Lyme Bay, Devon, in a fuel-efficient and timely manner. This also required superior manoeuvrability and a stable working platform for crew. To meet this latter specification, *Alysée* was fitted with three hydraulically operated davits along her port side, each supported by a 1tonne-capacity winch and star wheel haulers on fold-out arms. Another notable design feature was the creation of removable railings on the side, enabling a clear working area over the

vessel's side when required. The deck is complemented by a knuckle boom crane with a SWL of 1tonne. The davits, crane and special mussel farming equipment are powered by a single auxiliary Doosan PU066 engine, via twin hydraulic pumps.

Meanwhile, spacious storage lockers have been situated on both sides of the central anchor locker at *Alysée*'s forward end, with additional lockers provided to the front and side of the wheelhouse.

Waterjets proved the most reliable propulsive option, given that the vessel would be exposed to sections of water containing significant debris. To this end, the craft's two Cummins main engines feature a pair of sea water intakes apiece, for consistent performance. The Hamilton waterjets were supplied with that manufacturer's blue ARROW control system, for ease of use when completing tricky manoeuvring and berthing procedures, at low speed, within confined spaces and the restricted conditions of the typical offshore mussel farm.

At time of going to press, it was uncertain whether a second Wave Farmer 1400 vessel was in the pipeline; however, Alnmaritec has announced its intention to develop a "much larger platform" with the capacity to land as much as 50tonnes of mussels in a single trip.

### TECHNICAL PARTICULARS

Length, oa ..... 15.2m  
 Breadth, moulded ..... 5.3m  
 Displacement ..... 8.5tonnes (light)  
 Design, draught ..... 0.75m  
 Deck space (total) ..... 60m<sup>2</sup>  
 Deck capacity ..... 10tonnes/m<sup>2</sup>  
 Service speed ..... 12knots  
 Max speed ..... 20knots

Classification society ..... N/A  
 Other important international regulations complied with ..... MCA Cat 2  
 Main engine(s)  
     Make ..... Cummins  
     Model ..... QSC8.3  
     Number ..... 2  
     Output of each engine ..... 373kW  
 Gearbox(es)  
     Make ..... ZF  
     Model ..... ZF286 reversing gearbox  
     Number ..... 2  
     Output speed ..... 1.4:1  
 Waterjet(s)  
     Make ..... Hamilton HJ364  
         (with blue ARROW control system)  
     Number ..... 2  
 Deck machinery  
     1 x Palfinger PK8501MA hydraulic knuckle boom crane, 1tonne SWL  
     3 x North Sea Winches telescopic single-arm lifting davits, each with a 1tonne SWL winch  
     Owner-supplied hydraulic eqpt for mussel farm operation, incl: 1 drill rig, 1 seeder, 1 declumper, 1 grader, 1 cam grader  
 Bridge electronics  
     Radar(s) ..... Simrad 4G broadband radar scanner  
     Autopilot ..... Simrad AC 12  
     Other communications systems ..... Simrad  
         Rs 35 VHF  
         DSC radio  
     GPS ..... Simrad GS 25  
     Chart plotter ..... Simrad NSS12 multifunction touch display screen  
     Fire detection system ..... CO<sub>2</sub> flooding system  
 Onboard capacities  
     Fuel oil ..... 2,000litres  
     Fresh water ..... 215litres







## BENREOCH: CTV designed for smooth interactions with offshore platforms

|                                                         |                               |
|---------------------------------------------------------|-------------------------------|
| Builder .....                                           | <b>Veecraft Marine</b>        |
| Designer .....                                          | <b>Incat Crowther</b>         |
| Vessel's name .....                                     | <b>Benreoch</b>               |
| Owner/operator .....                                    | <b>Africa Diving Services</b> |
| Country .....                                           | <b>Nigeria</b>                |
| Flag .....                                              | <b>Nigeria</b>                |
| Total number of sister ships<br>already completed ..... | <b>1</b>                      |
| Total number of sister ships still on order .....       | <b>0</b>                      |
| Contract date .....                                     | <b>June 2012</b>              |
| Delivery date .....                                     | <b>May 2015</b>               |

South African shipyard Veecraft Marine extended its reach into the offshore marine sector in 2015 with the completion and delivery of a new wave-piercing crew transfer vessel (CTV) for Nigeria's African Diving Services. The Cape Town yard, which has previously specialised in the production of small commercial and naval vessels, rolled out the 30.3m loa *Benreoch* in late June, which has been designed – by Australia's Incat Crowther – to enable personnel transfers to offshore platforms located in Nigeria's offshore waters.

The contract has also seen Incat Crowther consolidate its influence in the African sector. The group notably drew up the plans for the 27.6m x 7.5m, 32knot, 2011-delivered *Topaz Zenith*, the first in its Topaz Z class of wave-piercing crew transfer catamarans, which was built at Nicocraft for Qatar-headquartered operator Topaz Marine. Incat Crowther has expanded on these dimensions to produce the arrangement for *Benreoch*, which, in addition to featuring an increased length, also extends the beam to 8.5m, complemented by a depth of 3.2m and a hull draught of 1.35m. Both the hull and the technician/passenger cabin are fashioned from marine grade aluminium.

The main beneficiary of these expanded dimensions is the CTV's aft working deck, which

provides space for a rescue boat, a crane and 53m<sup>2</sup> worth of cargo space, providing ample room for 10ft and 20ft containers, should the crew need to take supplies, equipment and gear to their offshore destinations. *Benreoch's* tanks enable the vessel to store up to 30,000litres of fuel oil, 4,000litres of fresh water and 500litres of sullage.

Special attention was paid to the design of the cat's bows, Incat Crowther reveals. The main consideration was to ensure that the vessel can "interface cleanly" with the offshore platform, providing a smooth connection for the onboard technicians to disembark. Both bows were subsequently positioned a suitable distance back from the outline of the foredeck, to protect the forepeaks and hull from damage, and to minimise the risk of the platform technicians sustaining crush-related injuries.

A galley and mess are located starboard, in addition to laundry facilities. *Benreoch* can accommodate up to 15 crew members and 22 technicians. The main deck features a cabin, which provides forward-facing seating for the platform technicians, as well as a captain's cabin and, for crew, two cabins, featuring four and six berths. The remaining crew resting space is located in the hulls, in the form of two twin cabins. Crew can access the main deck from the hulls via a flight of stairs, which also leads to the wheelhouse – fitted with anti-ballistics materials – on the upper deck.

*Benreoch* is powered by two Caterpillar C32 Acert main engines, each rated 1,081kW at 2,000-2,300rpm, driving twin propellers and providing a service speed of 26knots, increasing to 30knots max. Meanwhile, a pair of Caterpillar C4.4 generators provides the vessel's onboard electrical needs. The vessel has been flagged by Nigeria and classed by Bureau Veritas (BV), which has granted her BV \*HULL \*MACH, Crew Boat and Sea Area 2 notations.

### TECHNICAL PARTICULARS

|                                |                                            |
|--------------------------------|--------------------------------------------|
| Length, oa .....               | 30.35m                                     |
| Length, bp .....               | 27.31m                                     |
| Breadth, moulded .....         | 3.2m                                       |
| Gross tonnage .....            | 162tonnes                                  |
| Lightweight .....              | 77tonnes                                   |
| Deck space (total) .....       | 53m <sup>2</sup>                           |
| Deck capacity .....            | 2tonnes/m <sup>2</sup>                     |
| Service speed .....            | 28knots @ 85% MCR                          |
| Max speed .....                | 30knots                                    |
| Classification society .....   | Bureau Veritas                             |
| Notations .....                | * Hull * Machinery<br>Crew Boat Sea Area 2 |
| Main engine(s)                 |                                            |
| Make .....                     | Caterpillar                                |
| Model .....                    | Acert C32                                  |
| Number .....                   | 2                                          |
| Output of each engine .....    | 1,081kW @<br>2,300rpm                      |
| Gearbox(es)                    |                                            |
| Make .....                     | ZF                                         |
| Model .....                    | 3055                                       |
| Propeller(s)                   |                                            |
| Material .....                 | NiAlBr Cu-3                                |
| Fixed/controllable pitch ..... | Fixed                                      |
| Diameter .....                 | 1.118m                                     |
| Deck machinery ....            | 1 x crane, 3tonnes@4.4m                    |
| Bridge electronics             |                                            |
| Radar .....                    | 1 x Koden MDC-2510                         |
| Autopilot .....                | 1 x Navitron NT888G                        |
| GPS .....                      | 1 x Koden KGP913                           |
| Gyro .....                     | 1 x Sperry Navigation X Mk2                |
| Chart plotter .....            | 1 x Koden GTD150                           |
| Onboard capacities             |                                            |
| Fuel oil .....                 | 30,000litres                               |
| Fresh water .....              | 4,000litres                                |
| Sullage .....                  | 500litres                                  |
| Complement .....               |                                            |
| Crew .....                     | 15                                         |
| Passengers .....               | 22                                         |





# CLEAN HARBOUR 1 & CLEAN HARBOUR 2: Investigation of diesel-electric driven technology in Hong Kong

Builder's name ..... **Wang Tak Engineering & Shipbuilding Co., Ltd.**  
 Designer .... **Shanghai Chizhou Marine Tech. Ltd.**  
 Vessel's name ..... **Clean Harbour 1, Clean Harbour 2**  
 Owner/Operator ..... **Drainage Services Department, HKSAR**  
 Country ..... **China**  
 Flag ..... **Hong Kong**  
 Total number of sister ships already completed ..... **2**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **July 2013**  
 Delivery date ..... **March 2015**

The March 2015 launch of the first of two diesel-electric driven steel hull open-top container vessels *Clean Harbour 1* and *Clean Harbour 2* signified a bolster to Hong Kong's hybrid power ambitions. The vessels were delivered to Drainage Services Department (DSD), HKSAR by the builder Wang Tak Engineering & Shipbuilding Co., Ltd.

The vessels were designed as essential components for daily transportation of dewatered sludge cake from Stonecutters Island Sewage Treatment Works to the sludge treatment facility in Tuen Mun, under DSD Harbour Area Treatment Scheme Stage 2A. A total of 90 ISO 1C-type sludge containers can be carried inside each vessel's two cargo holds.

The sisters were designed by Shanghai Chizhou Marine Tech. Ltd, with diesel-electric propulsion systems designed by China Shipbuilding Industry Corporation Shanghai Marine Diesel Engine Research Institute 711. *Clean Harbour 1* and *Clean Harbour 2* were built and surveyed to Lloyd's Register rules and licensed as Hong Kong Marine Department Class IIA Dry Cargo Vessels. These are the first set of Hong Kong-licensed container vessels with diesel-electric propulsion for Hong Kong service.

Electrical propulsion consists of two CEEG CSD-900 transformers, two ABB ACS800-07-1060-7 variable frequency devices, two ABB HXR 450L G4 electric motors and two SJMATEK SZP75A azimuth thrusters. One SJMATEK SCT-90FP bow thruster was located at the bow for berthing assistance. The combination of azimuth thrusters and bow thruster

grants each vessel superior manoeuvrability and a small turning circle.

To suit Hong Kong's sea conditions and the operator's requirements, the vessels were designed with different combinations of genset outputs that can suit different operating conditions during voyages, in order to maximise efficiency of the gensets and reduce fuel consumption. Apart from manual control of the gensets, full automatic power management is performed by PMS. In emergency cases, the standby genset(s) can be synchronised and connected to the busbar automatically.

The engine control room consists of one control console, with an integrated monitoring system for gensets, thrusters and tanks. The wheelhouse consists of one console with main manoeuvring control system, navigation aids and radio communication. The deck rail-mounted IHI gantry crane can perform 20 loading and unloading container movements per hour. Crew accommodations, including single and double cabins, are provided at the forecabin deck, with the galley and toilet facility located on the main deck.

## TECHNICAL PARTICULARS

Length, oa ..... 69.9m  
 Length, bp ..... 67.36m  
 Breadth moulded ..... 17.49m  
 Depth, moulded ..... 5.5m  
 Gross tonnage ..... 2,190tonnes  
 Displacement ..... 3,400tonnes  
 Design, draught ..... 3.5m  
 Design, deadweight ..... 2,150tonnes  
 Lightweight ..... 1,250tonnes  
 Deck capacity (tonnes/m<sup>2</sup>) ..... 3.5tonnes/ m<sup>2</sup>  
 Speed, service (85% MCR output) ..... 10.5knots  
 Max speed ..... 11knots  
 Range (nautical miles) ..... 2,600nm  
 Fuel consumption ..... 0.45tonnes/2hr sailing journey (approx)  
 Classification society ..... Lloyd's Register  
 Notations ..... ✱100A1 container ship, hatch covers omitted in the hold, Hong Kong Water Service, ShipRight IHM, ✱LMC  
 Main engine(s)  
 Make ..... Caterpillar  
 Model ..... C32 /C18  
 Number ..... 2/ 1  
 Output of each engine ..... 2 x C32, 830kW each  
 1 x C18, 350kW

Propeller(s)  
 Material ..... Brass  
 Manufacturer ..... SJMATEK (Suzhou) Marine Machine Co  
 Number ..... 2  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1,650mm  
 Speed ..... 390rpm (approx)  
 Open or nozzled ..... Fitted  
 Bow thruster(s)  
 Make ..... SJMATEK (Suzhou) Marine Machine Co  
 Number ..... 1  
 Output ..... 200kW@1,500rpm  
 Azimuth thruster(s)  
 Make ..... SJMATEK (Suzhou) Marine Machine Co  
 Number ..... 2  
 Output of each ..... 750kW@1,500rpm  
 Deck machinery ..... 1 x IHI gantry crane, 20tonnes SWL  
 1 x Fo Shan Hai Xing Marine Machinery Co winch, drum diameter 36mm x 160m  
 5 x rollers, 147kN each  
 Other deck/machinery equipment ..... 2 x anchor windlass  
 2 x provision cranes (0.5tonnes / 0.9tonnes SWL)  
 Bridge electronics  
 Radar ..... Furuno 1945  
 GPS ..... Koden KGP-920  
 Chart plotter ..... Xinuo HM-9512-AIS  
 Engine monitoring / fire detection system ..... Jiaying Fucheng JBS-4  
 Onboard capacities  
 Fuel oil ..... 88.6tonnes  
 Fresh water ..... 39tonnes  
 Ballast water ..... 760.8tonnes  
 Complement  
 Crew ..... 12  
 Passengers ..... 0  
 Number of cabins ..... 8  
 Other significant or special items of equipment  
 ABB main switchboard, VFDs, electric motor  
 CEEG transformers



## CURTIS MARSHALL: Compact dive support vessel, optimised for offshore wind work

Builder ..... **B Marshall Marine Ltd**  
 Designer ..... **S C McAllister**  
 Vessel's name ..... **Curtis Marshall**  
 Owner / operator ..... **B Marshall Marine Ltd**  
 Country ..... **UK**  
 Flag ..... **UK**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **March 2015**

Formed in 2010 as part of a joint venture with the director of Sub Aqua Design Services, Middlesbrough, UK-based B Marshall Marine Ltd was created with one particular aim – the self-financing and construction of a vessel designed specifically for offshore diving support. According to B Marshall Marine, this sector has suffered from a distinct “lack of tailor-made dive support craft of the right size”, especially when it comes to activities related to offshore renewable energy, survey work and general repair and maintenance tasks.

This optimised-for-purpose vessel was realised in the form of *Curtis Marshall*, which left the blocks in March 2015, and which can be chartered from B Marshall Marine in association with Sub Aqua Diving Services. Despite her small dimensions, the vessel has been likened to ‘the Tardis’, Dr Who’s famous time-travelling police box, for the sheer amount of facilities she has managed to incorporate into her compact frame.

For instance, the 26m loa craft features: eight double berth cabins; a spacious, eight-seat galley; a utility area and a communal mess room; all enabling a high degree of comfort. This is an essential factor, given that *Curtis Marshall* has been designed to be capable of three weeks’ worth of self-sufficient sea time. The vessel’s safe and reliable working platform comprises: an onboard

decompression chamber; integrated dive control; the capability to transfer ballast water between either of the two aft tanks and the forward tank; and a four-point mooring system.

The craft is also equipped with a bespoke walkabout control system, which is detachable from the helm position and which can be used to control all four mooring winches, steering, propulsion and the bow thruster. With the offshore wind farm support sector in mind, the vessel has also been fitted with a customised bow fender, to protect the boat when it is pushing onto the wind turbine. The craft’s fender incorporates load cells for turbine load monitoring, for both the peace of mind of crew, and as a means of logging and producing data for client records. And, unlike many larger dive support vessels, *Curtis Marshall*’s compact nature provides the additional bonus of fuel economy; the vessel drinks a comparatively low rate of 120litres per hour.

### TECHNICAL PARTICULARS

Length, oa ..... 26m  
 Length, bp ..... 23.95m  
 Breadth, moulded ..... 7.7m  
 Depth, moulded ..... 2.6m  
 Gross tonnage ..... 110.97tonnes  
 Design, draught ..... 2.6m  
 Lightweight ..... 183tonnes  
 Deck space (total) ..... 105m<sup>2</sup>  
 Deck capacity ..... 30tonnes/m<sup>2</sup>  
 Service speed ..... 11knots@1,400% MCR  
 Max speed ..... 12.5knots  
 Range (nautical miles) ..... 1,000  
 Daily fuel consumption ..... 3tonnes  
 (for 24 hours steaming)  
 Bollard pull ..... 12tonnes  
 Classification society and notations ..... MCA  
 Cat 1 MGN 280  
 Main engine(s)  
 Make ..... Doosan

Model ..... 4V158TIH  
 Number ..... 2  
 Output ..... 395kW  
 Gearbox(es)  
 Make ..... Dong-I  
 Model ..... DMT 260TH  
 Number ..... 2  
 Output speed ..... 5.04:1 ratio  
 Propeller(s)  
 Material ..... Manganese  
 Manufacturer ..... Teignbridge  
 Number ..... 2  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1,500mm  
 Speed ..... 300rpm (max)  
 Open or nozzled ..... Open  
 Bow thruster(s)  
 Make ..... Kort Propulsion KT 120  
 Number ..... 1  
 Deck machinery  
 Crane(s) ..... 1 x ATLAS EK6500,  
 12.5tonnes SWL  
 Winch(es) ..... 4 x Hardy Engineering Ltd  
 winches, 7tonne capacity  
 Other deck machinery / equipment ..... 1 x  
 Rotzler deck winch, 25tonnes  
 Bridge electronics  
 Radar(s) ..... Furuno DRS 12A  
 Autopilot ..... Furuno Navipilot 700  
 GMDSS ..... Furuno  
 GPS ..... Furuno NavNet 3D  
 Chart plotter ..... Furuno NavNet 3D  
 Onboard capacities  
 Fuel oil ..... 13,000litres  
 Fresh water ..... 22,000litres  
 Sullage ..... 2,500litres  
 Complement  
 Crew ..... 4  
 Passengers ..... 12  
 Number of cabins ..... 8





## ERNST MEIER-HEDDE: New breed of German SAR and firefighting craft

Builder ..... **Fassmer**  
 Designer ..... **Lütten / Freese**  
 Vessel's name ..... **Ernst Meier-Hedde**  
 Owner/operator ..... **Deutsche Gesellschaft zur Rettung Schiffbrüchiger (DGzRS)**  
 Country ..... **Germany**  
 Flag ..... **German**  
 Total number of sister ships already completed ..... **0**  
 Total number of sister ships still on order ..... **2**  
 Contract date ..... **September 2013**  
 Delivery date ..... **May 2015**

*Ernst Meier-Hedde* is the first in a series of three new search and rescue (SAR) boats built by Fassmer for the German sea rescue service, the DGzRS, which is on a mission to rejuvenate its fleet.

Delivered in tandem with the DGzRS' 150th anniversary in 2015, the 27.9m x 6.2m newbuild, and her forthcoming two sisters, have been designed and constructed with increased length, power and enhanced, high-end navigational and communications equipment, and will replace three of the six existing 27.5m DGzRS SAR boats that were originally built for the service between 1985 and 1992.

The new vessel has been equipped with a comprehensive range of medical equipment, in addition to a fire pump, enabling the boat to tackle offshore or port-based blazes. *Ernst Meier-Hedde* also carries a daughter craft, named *Lotte*; she measures 8.2m x 2.9m, draws 0.8m and has a displacement of 4.3tonnes. Fitted with a single Steyr SE 236E40 engine, rated 170kW, *Lotte* can achieve a maximum speed of 19knots.

The main vessel deploys a knife-type towing hook, enabling her to partake in salvage operations,

if necessary. A pair of MTU engines drives a fixed propeller through ZF gearboxes, enabling the vessel to reach speeds of up to 24knots when called upon for first response work.

### TECHNICAL PARTICULARS

Length, oa ..... 27.9m  
 Length, bp ..... 25.7m  
 Breadth, moulded ..... 6.2m  
 Depth, moulded ..... 1.95m  
 Displacement ..... 120tonnes  
 Design, draught ..... 1.8m  
 Design, deadweight ..... 115tonnes  
 Lightweight ..... 97tonnes  
 Service speed ..... 24knots  
 Max speed ..... 24knots  
 Range (nautical miles) ..... 600@24knots  
 800@15knots

Classification society ..... Germanischer Lloyd  
 Notations ..... \*100 A5 HSDE 'Rescue Vessel'  
 Bollard pull ..... Uses knife-type towing hook, 7tonnes SWL

Main engine(s)  
 Make ..... MTU  
 Model ..... 16V 2000 M72  
 Number ..... 2  
 Output of each engine ..... 1,440kW

Gearbox(es)  
 Make ..... ZF  
 Model ..... ZF 5000 D  
 Number ..... 2  
 Output ..... 900rpm

Propeller(s)  
 Manufacturer ..... Schaffran  
 Number ..... 2  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1.1m  
 Speed ..... 900rpm  
 Open or nozzleed ..... Open (slightly tunnelled)

Alternator(s)  
 Make ..... Hansa Aggregate / Whisper Power  
 Number ..... 2  
 Output of each set ..... 88kVA / 26kVA / 1,500rpm

Bow thruster(s)  
 Make ..... Jastram  
 Number ..... 1  
 Output of each ..... 75kW

Deck machinery  
 2 x Steen anchor winches, plus capstan/wire aft, 2tonnes SWL each  
 Automatic daughter craft launch/retrieval system, via stern ramp/transom door

Bridge electronics  
 Radar(s) ..... Transas Naviradar 4000MFD  
 Furuno FAR 2117

Autopilot ..... Anschütz Pilotstar D  
 GMDSS ..... Sailor RT 6222 / RT 6248  
 Other communications systems ..... Rotheta RT-500-M

GPS ..... Saab / Furuno GP 150  
 Gyro ..... Anschütz Std 22  
 Chart plotter ..... 3 x Transas Navisailor 4000  
 Fire detection system ..... Böning

Onboard capacities  
 Fuel oil ..... 19m³  
 Fresh water ..... 4m³  
 Sludge ..... 1.95m³  
 Waste water ..... 2.5m³

Complement  
 Crew ..... 4  
 Passengers ..... 0  
 Number of cabins ..... 4

Other significant or special items of equipment  
 Daughter craft (*Lotte*)  
 Firefighting equipment: ..... 220m³/hr  
 Monitor, throwing range: ..... 80m  
 Portable salvage and bilge pumps





# GOBBLER 290: Revolutionary tank-free oil skimmer prototype

|                                                     |                                              |
|-----------------------------------------------------|----------------------------------------------|
| Builder.....                                        | <b>Gobbler Boats Ltd</b>                     |
| Designer ..                                         | <b>Gobbler Boats Ltd / Laurent Giles Ltd</b> |
| Vessel's name.....                                  | <b>Gobbler 290</b>                           |
| Owner/operator .....                                | <b>Gobbler Boats Ltd</b>                     |
| Country .....                                       | <b>UK</b>                                    |
| Flag .....                                          | <b>UK</b>                                    |
| Total number of sister ships already completed..... | <b>N/A</b>                                   |
| Total number of sister ships still on order .....   | <b>N/A</b>                                   |
| Contract date .....                                 | <b>N/A</b>                                   |
| Delivery date .....                                 | <b>N/A</b>                                   |

Gobbler Boats has unveiled the prototype version of its Gobbler 290 class of oil spill recovery craft, designed to be transported by road, air and vessel to spill sites in the quickest timeframes possible. Launched at Seawork 2015, the prototype of this class manages to combine compact and lightweight dimensions with complex and radical oil-skimming technology.

Made from fibreglass and incorporating oil-resistant resins, the Gobbler 290 weighs 3.5tonnes, draws 450mm and accommodates a two-man crew. Given these specs, the vessel / trailer combination can be launched easily from her custom-built, all-terrain road trailer to deal with spills directly from shore. Gobbler 290 vessels, sometimes including the trailer and truck, can also fit into the load space of various transport aircraft, for access to isolated areas. As such, the vessel can be put to use in a variety of remote and hard-to-access locations and can be launched in shallow waters. The boat's wheelhouse can also be detached and stowed on the aft deck in the event of limited aircraft headroom, or for passing under low road bridges.

The class also features a unique oil spill recovery set-up. The vessel can carry a range of oil skimmers for differing scenarios; one features a recovery capacity of 22tonnes per hour, while another is rated 40tonnes per hour, and both skimmer types deliver 98-100% water-free recovered oil. These skimmer types are located at the front of the vessel.

Throughout its oil recovery operations, the Gobbler 290 remains completely oil-free, save for the oil that it is actively recovering and handling. This recovered oil is pumped 'backwards', through the boat's internal hoses, to a bollard, located on the aft deck. The bollard is connected to a detachable bladder that is trailed behind the boat during her clean-up operations – typically conducted at a speed of 2knots. Each bladder has a capacity of up to 13,000litres and is infinitely reusable. Rather than having

to make regular trips back to shore to offload recovered oil, the Gobbler can fill each bladder to capacity, and then quick-release it from the vessel's aft for later collection, while applying a new, empty bladder to the deck transfer bollard.

To support these operations, Gobbler Boats has also devised and produced a series of floating 'bladder collection points', in the form of the Quad-Pod and the Octo-Pod. Each features a light and AIS beacon, and is used to store spare bladders (eight on the Quad-Pod / 16 on the Octo-Pod, including two spares for each group-working vessel) The pods are weight-matched to the lifting capacities of various helicopters and can also be deployed by an accompanying mothership. Each pod also incorporates a refuelling facility. When the Gobbler 290 crew has filled one bladder, they can connect it to the pod and then collect a spare bladder to continue their work. When the pods' bladders have all been filled, they can be simultaneously pumped, via the pod central pumping system, to an accompanying mothership, and the empty bladders are then ready for re-use, to reduce fuel consumption and save time.

The Gobbler 290 incorporates solid state technology and hydraulically controlled equipment throughout, resulting in a "better than 95% spark-free environment", Gobbler Boats Ltd confirms. In the event of a flameout, the crew can detach the bladder from the bollard and make a speedy getaway from the scene.

The Gobbler 290 is also fitted with a jet intake auto-clean system so that, should the waterjet intake screen encounter an obstruction, the crew can pull a lever connected to the vessel's PRM gearbox and blast the object out backwards. This saves the crew from having to dive underneath the boat to manually clear the underwater obstruction, and protects them from potential sub-zero temperatures and attacks by marine wildlife.

## TECHNICAL PARTICULARS

|                          |                         |
|--------------------------|-------------------------|
| Length, oa .....         | 8.85m                   |
| Length, bp .....         | 8m                      |
| Breadth, moulded.....    | 2.55m                   |
| Gross tonnage.....       | 3.75tonnes              |
| Displacement .....       | 3.75tonnes              |
| Design, draught .....    | 450mm                   |
| Design, deadweight ..... | 3.75tonnes              |
| Lightweight .....        | 3.25tonnes              |
| Deck space (total).....  | 10m² (approx)           |
| Deck capacity .....      | N/A (work area only)    |
| Service speed .....      | 2knots (while skimming) |

|                              |                                    |
|------------------------------|------------------------------------|
| Max speed .....              | 23knots                            |
| Range (nautical miles) ..... | 150                                |
| Daily fuel consumption ..... | 400litres                          |
| Classification society ..... | Lloyd's Register                   |
| Notations.....               | ✱100A1 SSC Workboat<br>Mono HSC G3 |

|                                                               |                                                                                              |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Other important international regulations complied with ..... | USCG<br>Lloyd's structural 150nm offshore<br>Complies for installation on commercial vessels |
| Bollard pull.....                                             | 0.5tonnes                                                                                    |
| Main engine(s) .....                                          |                                                                                              |
| Make .....                                                    | FNM (Steyr/Buch, depending on operational scenarios)                                         |
| Model .....                                                   | 300b                                                                                         |
| Number .....                                                  | 1                                                                                            |
| Output of each engine.....                                    | 184kW+                                                                                       |
| Gearbox(es) .....                                             |                                                                                              |
| Make .....                                                    | PRM Newage                                                                                   |
| Model .....                                                   | 500/750 (depending on engine)                                                                |
| Number .....                                                  | 1                                                                                            |
| Output speed .....                                            | 1.5:1<br>(2:1, depending on engine)                                                          |

|                                    |                                          |
|------------------------------------|------------------------------------------|
| Alternator(s) .....                |                                          |
| Make/type .....                    | Bosch                                    |
| Number .....                       | 1                                        |
| Output/speed .....                 | 150A                                     |
| Waterjet(s) .....                  |                                          |
| Make....                           | Hamilton 247 / NamJet / AlmarineJet 288  |
| Number .....                       | 1                                        |
| Deck machinery.....                | 1 x Maxwell hydraulic winch              |
| Bridge electronics .....           |                                          |
| Radar(s) .....                     | Garmin                                   |
| Autopilot .....                    | Simrad AP60                              |
| GMDSS .....                        | B&G V50 DSC/VHF                          |
| Other communications systems ..... | EPIRB                                    |
| GPS .....                          | GPSMAP 1020XS                            |
| Chart plotter .....                | GPSMAP 1020XS MOB                        |
| Fire detection system.....         | Seafire 4-circuit engine shutdown system |

|                          |           |
|--------------------------|-----------|
| Onboard capacities ..... |           |
| Fuel oil.....            | 400litres |
| Fresh water .....        | bottled   |
| Complement .....         |           |
| Crew .....               | 2         |
| Passengers.....          | 7         |
| Number of cabins .....   | 0         |





## GUSTAF AF KLINT: Unique 'clear sweeping' vessel for the Swedish Maritime Administration

Builder ..... **Dockstarvarvet**  
 Designer ..... **Petter Hakanson Marine /DV**  
 Vessel's name ..... **Gustav AF Klint**  
 Owner/operator ..... **Swedish Maritime Administration (SMA)**  
 Country ..... **Sweden**  
 Flag ..... **Sweden**  
 Total number of sister ships  
   already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **November 2013**  
 Delivery date ..... **June 2015**

The Dockstarvarvet-built *Gustaf AF Klint* does not so much represent a 'first-in-class' vessel, but more a "specialised, one-off project", in the words of the builder's CEO, Anders Hellman. However, Hellman admitted, it was a job which has proven an ideal crash course in "system integration and electric drive installations," which the yard can carry over into future orders.

Named after a pioneer in Swedish hydrographic survey, who was active between 1790-1840, *Gustaf AF Klint* has been designed to specialise in 'clear sweeping' work, an offshore activity not entirely new to Dockstarvarvet; the yard built three clear sweeping ships in the 1970s. This necessitates the vessel carrying a 30m steel beam – hinged in a winch system, sideways-on, across the vessel's stern – which is moved at a slow speed, at a pre-set depth, and which registers any contact with objects on the seafloor. This is used for tasks where the demand for accuracy cannot be met with conventional multi-beam echo-sounding technology – including areas such as fairways and harbour waters.

The ship is a trimaran, fashioned purely from aluminium, and utilises a sole Volvo-Penta diesel engine, rated 558kW, and Twin Disc MGX-5204 gearbox with trolling function, generating only 25rpm on the propeller in dead slow mode. Meanwhile, the vessel's forward and aft thrusters enable the ship to sail on a straight line, independent of wind and current, up to certain limits.

Although most sweeping operations will be conducted when going forward, they will sometimes be conducted when going astern – for instance, when working against a dock wall. The sweep system is controlled via a mounted platform on the bridge, complete with two crew seats. The platform is rotatable; when working in astern mode, for example, the platform is turned 180deg.

*Gustaf AF Klint* has also been fitted with a comprehensive electronics package, enabling the crew to register signals from the sensors on the steel beam and the beam winches, as well as positional and depth data. Altogether, about 50 sensors and antennas have been installed aboard the vessel, as have some 20 computers, with presentations accessible via 10 monitors.

The main deck level of the deckhouse contains a workshop and a stores room for survey equipment, as well as mess, toilet/washroom and berth facilities for the crew. An A-frame, capable of lifting 1,500kg, is fitted on the aft deck for handling of the workboat, a ROV and/or a 10ft container. The vessel's windlass, A-frame, sensor system and winch system for the beam, as well as the entire control system for main engine and thrusters, were designed and delivered by Petters Marine Hydraulic (PMH). The system contains functions for six different modes: conventional manoeuvring with rudder and propellers; joystick mode; keep position (fixed on one spot); 'slow speed track' (while sweeping at speeds up to 2knots); 'high speed track' in speeds of up to 8knots, for multi-beam sounding work; and transit mode on auto-pilot. The sensor and manoeuvring systems have also been integrated with the SMA's shore-based systems for position fixing and registration of depth data.

### TECHNICAL PARTICULARS

Length, oa ..... 17.6m  
 Length, bp ..... 17.22m  
 Breadth, moulded ..... 8.6m  
 Depth, moulded ..... 2.7m  
 Gross tonnage ..... 78tonnes  
 Displacement ..... 54.24tonnes  
 Design, draught ..... 1.5m  
 Design, deadweight ..... 6.24tonnes  
 Lightweight ..... 48tonnes  
 Deck space (total) ..... 40m²  
 Service speed ..... 0-2knots (when bar swaying)  
 Max speed ..... 12knots  
 Range (nautical miles) ..... 200  
 Daily fuel consumption ..... 110litres  
 Classification society ..... Swedish Transport Agency  
 Notations ..... Area C  
 Main engine(s)  
   Make ..... Volvo Penta  
   Model ..... D16

Number ..... 1  
 Output of each engine ..... 551kW  
 Gearbox(es)  
   Make ..... Twin Disc  
   Model ..... MGX-5204  
   Number ..... 1  
   Output ..... 900rpm  
 Propeller(s)  
   Material ..... NiBrAl  
   Manufacturer ..... Teignbridge  
   Number ..... 1  
   Fixed/controllable pitch ..... Fixed  
   Speed ..... 900rpm  
   Open or nozzled ..... Open  
 Alternator(s)  
   Make ..... Volvo Penta D9 MG HE/KC  
   Number ..... 1  
   Output of each set ..... 281kVA @ 1,500rpm  
 Bow thruster(s)  
   Make ..... Sleipner  
   Number ..... 1  
   Output of each ..... 71kW  
 Stern thruster(s)  
   Make ..... Sleipner  
   Number ..... 2  
   Output of each ..... 38kW  
 Deck machinery  
   1 x MB Hydraulikk anchor winch  
   1 x PMH frame  
   6 x PMHG bar sweeping systems, capacities of 2x1,500kg plus special winches  
 Bridge electronics  
   Radar ..... Furuno RSB-118  
   Autopilot ..... PMH Keep Position system  
   GMDSS ..... Furuno / McMurdo  
   Other communications systems ..... Furuno  
   GPS ..... SMA Special  
   GPS compass ..... Furuno SC-30-BB  
   Chart plotter ..... Furuno / MaxSea  
   AIS ..... Saab R4 inland  
   Engine monitoring system ..... Volvo Penta / Twin Disc  
 Fire detection system ..... Fire Pro  
 Onboard capacities  
   Fuel oil ..... 2,260litres  
   Fresh water ..... 1,000litres  
   Sullage ..... 500litres  
 Complement  
   Crew ..... 2+2  
   Passengers ..... 0  
   Number of cabins ..... 2



## HELIOTROPE 48: Pleasure cat built to blend high comfort with high performance

Builder ..... **Bakricono Shipyard**  
 Designer ..... **Albatross Marine Design**  
 Vessel's name ..... **Heliotrope 48**  
 Owner/ operator ..... **Undisclosed**  
 Country ..... **Thailand**  
 Flag ..... **Thailand**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **2**  
 Contract date ..... **September 2014**  
 Delivery date ..... **September 2015**

Designed and built in Thailand – by Albatross Marine Design (AMD) and Bakricono Shipyard, respectively – the concept behind the new Heliotrope 48 (H48) class is to match eye-catching shapes and striking colours with practical functionality and operational efficiency, so as to create a genuinely unique entrant to the power catamaran yacht market. This market has undergone a surge of popularity in recent years, and remains a potential growth area for yachting enthusiasts who may not be able to afford the ‘full trimmings’ of a luxury yacht, but whose wallets can stretch to a quality recreational vessel that does not require the presence of permanent crew, nor extensive mooring fees.

However, AMD has been keen to ensure that the H48 is not mistaken for a ‘marina-based houseboat’, but instead fulfils her role as a performance-oriented model. The cat’s hull form was developed following detailed resistance studies, incorporating CFD methods, to provide fuel efficiency and soft rides across a wide range of semi-planing speeds. Special attention was also paid to the bridge deck configuration, as Dr Albert Nazarov, AMD managing director, recounts. “On many pleasure cats, the bunks are located at the bridge [or wet] deck level, commonly between the demi-hulls,” he explains.

“This means the designers have to reduce bridge deck height above waterline, often compromising the seakeeping.

“With the H48, we did not place any beds on the bridge deck. As a result, we can raise 25% of the forward area of the bridge deck, thus improving the seakeeping and reducing possible slamming on the seaway.” Additionally, the cat’s propellers and rudders are protected by skegs, facilitating beaching of the vessel.

Vacuum infusion was used to assemble the H48’s hull structure, drawing on a mixture of composites such as vinyl ester resin, multiaxial E-glass fabrics and Airex foam core. Below waterline, Lantor Soric bulking mats have been applied to the hull, and the cross structure has been reinforced, with unidirectional tape, in areas prone to slamming. This approach has succeeded in keeping overall weight down, whilst resulting in a sturdy and reliable fast vessel.

The H48 can be configured for different requirements. For instance, the ‘charter’ version of the craft offers four double cabins and four toilets, with two shared shower rooms; the ‘master’ option, meanwhile, can be made to feature a cabin and bathroom on one side of the vessel, with two guest blocks on the other. Featuring a 22m<sup>2</sup> flybridge, which opens up into the saloon to form a single functional zone, and with interior headroom pushed up to 2.2m, the vessel offers ample space for a variety of onboard activities.

### TECHNICAL PARTICULARS

Length, oa ..... 14.75m  
 Length, bp ..... 14.75m  
 Breadth, moulded ..... 7.23m  
 Depth, moulded ..... 2.59m  
 Displacement ..... 24tonnes  
 Design, draught ..... 1.1m

Design, deadweight ..... 5tonnes  
 Lightweight ..... 19tonnes  
 Service speed ..... 12knots @ 70% MCR  
 Max speed ..... 23knots  
 Range (nautical miles) ..... 500 (@10knots)  
 Classification society ..... IMCI  
 Notations ..... Pleasure craft EU RCD, Cat B  
 Other important international regulations  
 complied with  
 ISO Small Craft Standards  
 Main engine(s)  
 Make ..... Steyr  
 Model ..... SE266S36  
 Number ..... 2  
 Output of each engine ..... 190kW  
 Gearbox(es)  
 Make ..... ZF  
 Model ..... ZF63IV  
 Number ..... 2  
 Propeller(s)  
 Manufacturer ..... Teignbridge  
 Number ..... 2  
 Fixed / controllable pitch ..... Fixed  
 Diameter ..... 0.6m  
 Generator  
 Make/type ..... Fischer Panda 15000i  
 Number ..... 1  
 Output of each set ..... 12kW  
 Deck machinery ..... 1 x Maxwell anchor windlass  
 Bridge electronics  
 Radar(s) ..... Simrad  
 Onboard capacities  
 Fuel oil ..... 2,000litres  
 Fresh water ..... 800litres  
 Sullage ..... 600litres  
 Complement  
 Crew ..... 2  
 Passengers ..... 10  
 Number of cabins ..... 4











## KAROLINE: First battery-powered fishing boat curbs noise, fumes and vibration

Builder ..... **Selfa Arctic**  
 Designer ..... **Selfa Arctic**  
 Vessel's name ..... **Karoline**  
 Owner/operator ..... **Ora AS**  
 Country ..... **Norway**  
 Flag ..... **Norway**  
 Total number of sister ships  
   already completed ..... **1**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **October 2015**

*Karoline* has been hailed as the world's first battery-driven fishing vessel, developed by Norwegian boatbuilder Selfa Arctic. The builder specialises in the production of fishing vessels fashioned from fibreglass, via vacuum infusion, and is, at present, the only Norwegian boatbuilder entitled to approve its own newbuilds, as decided by the Norwegian Maritime Authority.

In 2007, Selfa Arctic initiated a project with the aim of developing environmentally friendly products and contributing to the global, industry-wide effort to curb vessel emissions. In collaboration with SINTEF and FHF (the former being Scandinavia's largest independent research organisation; the latter being The Norwegian Seafood Research Fund) a 'pre-project' was initiated in 2009. The result was a base calculation of the energy needed for a hybrid fishing vessel to carry out operations.

Based on the results of this study, Selfa Arctic then began to search for a suitable European component supplier, and entered into cooperation with Siemens in Trondheim, located 1.5km from Selfa's boatyard. The main project commenced in 2013 and was partially funded by Transnova (now Enova) and the Norwegian Research Council.

*Karoline* has been equipped with a Volvo Penta D4-180 generator, which drives a Siemens electric motor,

generating an output of 60kW at 2,400rpm. This also charges a 195kW battery bank, supplied by Corvus. As a result, the boat can carry out typical fishing vessel operations on battery power, which, for the fisherman, means drastically reduced noise, vibrations and toxic fumes. Steaming in and out can either be conducted via battery, or with the Siemens motor. The boat can be charged by a standard 63A cable when docked, with charging times estimated to last between six and eight hours.

During 2014 and 2015, the propulsion system was built on the floor at the Siemens plant in Trondheim and tested during the winter. The propulsion system and battery were subsequently installed in spring 2015. The boat was put to sea at the beginning of August and presented at Aqua Nor and then put through its paces in a 600nm sea trial, on a route between Trondheim and Tromsø.

### TECHNICAL PARTICULARS

Length, oa ..... 10.99m  
 Length, bp ..... 10.65m  
 Breadth, moulded ..... 4.2m  
 Depth, moulded ..... 1.99m  
 Displacement ..... 38tonnes  
 Design, draught ..... 0.75m  
 Lightweight ..... 16tonnes  
 Deck space (total) ..... 24m<sup>2</sup>  
 Service speed ..... 8.5knots @100% MCR  
 Range (nautical miles) ..... 260  
 Fuel consumption ..... 17litres/hr  
                                           w/ generator @ 60kW  
 Classification society ..... None  
 Other important international regulations  
   complied with ..... Nordic Boat Standard  
                                           NMA rules for fishing  
                                           vessels below 15m loa  
 Main engine(s)  
   Make ..... Siemens  
   Number ..... 1

Output of each engine ..... 60kW  
 Gearbox(es)  
   Make ..... ZF  
   Model ..... 280-1  
   Number ..... 1  
 Propeller(s)  
   Material ..... Bronze  
   Number ..... 1  
   Speed ..... 1,333rpm  
   Open or nozzled ..... Open  
 Bow thruster(s)  
   Make ..... Sleipner Motors  
   Number ..... 1  
 Stern thruster(s)  
   Make ..... Sleipner Motors  
   Number ..... 1  
 Deck machinery  
   1 x Selfa crane, 0.5tonnes SWL  
   1 x Lorentzen Hydraulikk net hauler, 1,200kg  
   1 x Net-Op net clearer  
   1 x Lorentzen Hydraulikk bleeding tank  
   1 x Lorentzen Hydraulikk arm for net clearer  
 Bridge electronics  
   Radar(s) ..... Simrad 4G18  
   Autopilot ..... Simrad AP70  
   GMDSS ..... McMurdo  
   GPS ..... JRC  
   Chart plotter ..... Olex  
   Fire detection system ..... Servofighter  
 Onboard capacities  
   Fresh water ..... 600litres  
   Ballast water ..... 800litres  
   Diesel oil ..... 500litres  
 Complement  
   Crew ..... 1  
   Passengers ..... 0  
   Number of cabins ..... 0  
 Other significant or special items of equipment  
   Series hybrid propulsion system



## L'ESCAPADE: Dinner for 60 on one of France's latest hybrid power passenger boats

Builder ..... **Alumarine Shipyard**  
 Designer ..... **DN&T**  
 Vessel ..... **L'escapade**  
 Owner/operator ..... **Communauté de Communes des 2 Vallées de France**  
 Country ..... **France**  
 Flag ..... **France**  
 Total number of sister ships already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **August 2015**

Emerging in the wake of the European marine sector's current 'electric renaissance', *L'escapade* joins the many passenger vessels within this territory to have selected hybrid diesel-electric power for operations. The vessel's construction has also involved what builder Alumarine terms a "pooling of European know-hows", with the vessel's diesel engine manufactured in Sweden, her battery management system sourced from the UK, her naval architecture provided by Belgium, her batteries delivered from Germany and her hybrid control system made in Italy.

The 27.8m x 5.05m vessel, which is capable of accommodating up to 102 passengers at sea – or 60 persons whilst undertaking an onboard restaurant service – has been fitted with a Volvo D5 engine, rated 118kW at 2,300rpm, linked to a

Transfluid HM2000 hybrid transmission and a Transfluid Powershift Revermatic 11-700RBD drive, featuring a reduction ratio of 1.88. The electric engine is rated 75kW and derives its power courtesy of a 150kW lithium phosphate battery pack, supplied by Exide FeLiPO, and enables the operator to switch between diesel and electric modes; the latter mode not only curbing emissions and fumes, but also for drastically reducing noise, leading to silent river operations and a noise level of no more than 58dB within the passenger area. This factor was a prime stipulation for her operations in the picturesque province of Oise, France.

In diesel mode, the vessel can attain a top speed of 10knots and a cruising speed of 7.5knots at 1,800rpm. In electric mode, the cruising speed would be in the region of 4.3knots, with the electric engine performing at 900rpm, increasing to 7.5knots at 1,800rpm max. Vessel trials proved *L'escapade* to have a range of 20 hours when operating at 5.4knots in pure electric mode.

The boat has been fashioned from aluminium, with operational flexibility built in; for instance, the passenger deck can be converted into a restaurant and reception room. The aft deck features a functional catering zone, and an outside folding roof terrace can accommodate up to 20 people. Passengers can stow their bicycles at a dedicated area at the bow.

In all, the launch and delivery of *L'escapade* adds to the many deliveries this year (and over

the past five years) that have proven that the rise of green craft technology is no flash-in-the-pan fad, and that it is possible to run a commercial passenger service, at optimum river cruising speed, whilst reducing emissions, noise and fuel oil consumption.

### TECHNICAL PARTICULARS

Length, oa ..... 27.8m  
 Length, wl ..... 24.7m  
 Breadth, moulded ..... 5.05m  
 Depth, moulded ..... 1.7m  
 Displacement ..... 34.5tonnes  
 Design, draught ..... 0.77m  
 Service speed ..... 3.8-7knots  
 Max speed ..... 10knots (diesel mode)  
 Classification society ..... Bureau Veritas (hull)  
 Other important international regulations complied with ..... French inland waterways standards

Main engine(s)  
 Make ..... Volvo Penta / Transfluid  
 Model D5 (diesel) / HM2000 (electric engine)  
 Number ..... 2  
 Output of each engine ..... 117.7kW / 75kW

Onboard capacities  
 Fuel oil ..... 1,800litres  
 Fresh water ..... 1,000litres  
 Complement  
 Passengers ..... 102





# LONDON TITAN: Multipurpose solution for River Thames maintenance, support and dredging work

Builder ..... **Manor Marine**  
 Designer ..... **Macduff Ship Design**  
 Vessel's name ..... **London Titan**  
 Owner/operator ..... **Port of London Authority**  
 Country ..... **UK**  
 Flag ..... **UK**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **January 2012**  
 Delivery date ..... **June 2015**

## TECHNICAL PARTICULARS

Length,oa ..... 36.5m  
 Length,bp ..... 34m  
 Breadth, moulded ..... 13.5m  
 Depth, moulded ..... 3.5m  
 Gross tonnage ..... 428tonnes  
 Displacement ..... 910tonnes (max)  
 Design, draught ..... 2.2m  
 Lightweight ..... 612tonnes  
 Deck space (total) ..... 200m<sup>2</sup>  
 Max speed ..... 12knots  
 Bollard pull ..... 30tonnes  
 Classification society ..... Lloyd's Register  
 Notations .... \*100A1 MCH Unrestricted Coastal  
 Area Navigation

Main engine(s)  
 Make ..... Caterpillar  
 Model ..... C32  
 Number ..... 3  
 Output of each ..... 745.7kW  
 Gearbox(es)  
 Make ..... Twin Disc  
 Model ..... MGX 5321  
 Number ..... 3  
 Propellers  
 Material ..... NiAlBr  
 Manufacturer ..... Kort  
 Number ..... 3  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1,500mm  
 Open or nozzled ..... Nozzled (Kort XL profile)

Alternators  
 1 x 69kW 400v 3ph 50hz driven by Aux 1  
 1 x 69kW 400v 3ph 50hz driven by Aux 2  
 3 x 75A 24v DC on main engines  
 1 x 60A 24v driven by Aux1  
 1 x 60A 24v DC driven by Aux2

Bow thruster(s)  
 Make/model ..... Kort KT 150  
 Number ..... 1  
 Output ..... 105kW

Deck machinery  
 Crane(s)  
 Manufacturer ..... SMT KTB  
 Number ..... 2  
 Capacities of each ..... 11tonnes@20.1m SWL

Winch(es)  
 Manufacturer ..... North Sea  
 Number ..... 9  
 Capacities of each ..... 1 x GM-600K anchor  
 handling winch, 120tonnes@4m/min  
 in high displacement of 1st layer  
 or 60tonnes at 8m/min in low  
 displacement on 1st layer.  
 Capacity for 100m of  
 4mm wire in three layers

1 x DM-550 anchor handling  
 winch, 20tonnes@ 20m/min  
 in high displacement on

1st layer or 10tonnes@40m/min  
 in low displacement on  
 1st layer. Capacity for 100m of  
 32mm wire in three layers

1 x GM/AW 50-22 hydraulic  
 anchor/mooring winch.

Duty pull 2,145kg, overload  
 pull 3,217kg at 11m/min  
 on chain gypsy. 3.3tonnes  
 @14m/min on mooring drum

1st layer. 2.2tonnes at  
 2.1m/min on mooring  
 drum 4th layer. Capacity for 67m  
 of 32m diameter rope in four layers

1 x CW-200 hydraulic gilsen winch,  
 19.5tonnes @13.3m/min on 1st layer,  
 100m of 32mm wire rope

4 x CW-100 hydraulic gilsen winch,  
 10tonnes @17m/min on 1st layer.  
 Capacity for 3,000m of 18mm wire rope

1 x CW-50 hydraulic gilsen winch,  
 5tonnes@20m/min on 1st layer,  
 speed 29rpm. Capacity for  
 70m of 20mm wire rope

Roller(s)  
 Number ..... 2 (1 bow, 1 stern)  
 Capacities ..... recessed 1m dia, SWL 120tonnes

Other deck/machinery equipment  
 3 x North Sea CH-60 hydraulic  
 capstans having a normal  
 running pull of 6tonnes at 20m/min  
 1 x North Sea gn-70 storage reel,  
 rated 1.8tonnes on 1st layer /  
 1.1tonnes on top layer

Bridge electronics  
 Radar ..... JRC JMA  
 Autopilot ..... Simrad AP50  
 GMDSS ..... 2 x Icom IC-GM1600  
 GPS ..... KGP-913  
 Gyro ..... JRC JLR-20  
 Chart plotter ..... Navi Sailor 4000  
 Other communications systems

JRC NCR333 Navtex receiver  
 2 x Sailor 6222  
 DSC VHF Class A  
 Transas Class B AIS transceiver  
 Simrad SA50 SART  
 Simrad RP50 EPIRB

Onboard capacities  
 Fuel oil ..... 72,000litres  
 Fresh water ..... 8,000litres  
 Ballast water ..... 167,000litres

Complement  
 Crew ..... 6  
 Passengers ..... 0  
 Number of cabins ..... 0

Described as a 'maid of all work' and a floating 'Swiss Army Knife', due to her versatility when it comes to riverine operations, the UK Port of London Authority's (PLA's) maintenance vessel *London Titan*, delivered in June 2015, represents the authority's single largest investment in 20 years, boasting a value of £6.9 million (US\$11.5 million).

*London Titan's* rota of tasks includes, but is not restricted to, mooring maintenance, buoy laying, dive support, pile removal, plough dredging and wreckage removal across the PLA's 95-mile jurisdiction on the River Thames. Designed by Aberdeenshire-based naval architect Macduff Ship Design and constructed by Dorset builder Manor Marine, the 36.5m loa *London Titan* has effectively replaced two 1960s-built PLA craft, *Hookness* and *Crossness*, resulting in a modernised vessel with reduced overall operating costs.

The original contract for the vessel stems back to January 2012, reflecting the attention that Macduff Ship Design and the PLA devoted to getting the vessel exactly right for her environment. Practical considerations included the creation of a shallow draught (the completed vessel has attained a maximum draught of 2.2m) and relatively 'squat' shape, so as to enable the vessel to pass under the many bridges encountered on the Thames, albeit with the capability to operate efficiently in open water stretches. Additionally, Macduff's team spent time aboard the outgoing duo *Hookness* and *Crossness*, to create as ergonomic an interior environment as possible for the crew.

Additional features aboard *London Titan* include: a pair of large articulated cranes; spud legs, located forward and aft of the starboard / port cranes, respectively, and installed as a less time-consuming alternative to anchors; a pair of large articulated cranes; and superior winch capacity to her two predecessors. A towed plough dredge, deployed over the stern by a gantry, enables the vessel to undertake dredging duties.

Consilium UK has designed and installed *London Titan's* full bridge navigation package, which it did to meet international shipping regulations for vessels of more than 400gt – another rarity for vessels navigating the River Thames, and a feature that could see the Lloyd's Register-classed ship deployed in other locations/regions if necessary.



## MAPLIN: First hydrographic survey vessel from CTruk, soon to make Thames debut

Builder ..... **CTruk Boats**  
 Designer ..... **CTruk Boats**  
 Vessel's name ..... **Maplin**  
 Owner/operator ..... **Port of London Authority**  
 Country ..... **UK**  
 Flag ..... **UK**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **January 2016**

UK builder CTruk continues to transplant the knowledge it has accumulated in offshore wind farm support vessel design and construction into new avenues.

Just two years after making its *Significant Small Ships* debut (in 2011) with the 18.5m crew transfer vessel *CWind Athenia*, the builder featured again in the journal with the completion of its prototype Twin Hulled Offshore Raider (THOR) model. This year, the group has stepped up to the plate with the January 2016 delivery of its first hydrographic survey boat to date.

Constructed for the Port of London Authority (PLA), the 17m x 6.1m *Maplin* is based on CTruk's MPC19 catamaran class, which has won plaudits for its seakeeping characteristics and notable levels of stability within the offshore wind farm services sector. Marrying corrosion-free composite design to a catamaran hull form, the vessel is intended to serve as a low-maintenance and relatively lightweight survey vessel, albeit one with sufficient deck space to enable a wide range of onboard activities.

This enhanced stability is particularly intended to increase *Maplin's* surveying capability; for

instance, in allowing more effective use of the multi-beam deployable sonar unit that has been built into the hull, CTruk says.

The deck area, strengthened to accommodate loads of up to 1tonne per m<sup>2</sup>, has been fitted with multiple fixing points, which enable the crew to lash and carry additional storage – for instance, small buoys, lifted aboard by the vessel's Bonfiglioli 2300 crane.

The wheelhouse has also been built with a low profile, and *Maplin's* mast made retractable, to facilitate the vessel's passage through waterways subject to low bridges. Her low draught will enable the vessel to pass under London Bridge.

*Maplin* has also been designed with a RIB recovery A-frame, thereby enabling the crew to launch and recover a 4m daughter craft. She is set to replace the PLA's 1955-built survey boat *Yantlet*, which is reaching retirement.

### TECHNICAL PARTICULARS

Length, oa ..... 17m  
 Length, wl ..... 15.8m  
 Breadth, moulded ..... 6.1m  
 Depth, moulded ..... 0.7m (light)  
 ..... 1.2m (max)  
 Gross tonnage ..... 26.5tonnes  
 Displacement ..... 28tonnes  
 Deck space (total) ..... 6m x 5.5m  
 Service speed ..... 18knots  
 Max speed ..... 24knots  
 Classification society ..... Bureau Veritas  
 Notations ..... Hull windfarm service 1, Sea Area 3  
 Other important international  
 regulations complied with .... MCA MGN280 Cat 2  
 Main engine(s)  
 Make ..... Volvo Penta  
 Model ..... DI3-700

Number ..... 2  
 Output of each engine ..... 515kW  
 Gearbox(es)  
 Make ..... Twin Disc  
 Model ..... MG – 5I35 – SC  
 Number ..... 2  
 Ratio ..... 1.28:1  
 Waterjet(s)  
 Make ..... Rolls-Royce  
 Model ..... FF41  
 Number ..... 2  
 Deck machinery  
 1 x Bonfiglioli 2300 crane, 0.72tonnes@3m /  
 0.37tonnes@5.2m SWL  
 1 x Hercules Hydraulics fixed winch, max  
 capacity 1.2tonnes  
 1 x CMAX portable sonar winch  
 1 x RIB launch hydraulic A-frame,  
 SWL 0.5tonnes  
 1 x electric lowering mast  
 Bridge electronics  
 Radar(s) ..... Simrad 6kW open array  
 Autopilot ..... Autopilot AC70  
 Other communications systems ..... Saab AIS  
 ..... 2 x Icon VHF  
 GPS ..... Simrad NSO EVO 2  
 Gyro ..... HS70 GPS compass  
 Engine monitoring system ..... Stat-x  
 Fire detection system ..... ED 820  
 Onboard capacities  
 Fuel oil ..... 5,000litres  
 Fresh water ..... 340litres  
 Sullage ..... 340litres  
 Ballast water ..... 2,000litres  
 Complement  
 Crew ..... 2  
 Passengers ..... 10  
 Number of cabins ..... 0





## NEIL ARMSTRONG: new oceanographic survey vessel developed and future-proofed for a wide array of missions

Builder's name: ..... **Dakota Creek Industries**  
 Designer: .... **Guido Perla & Associates (GPA)**  
 Vessel's name: ..... **Neil Armstrong**  
 Owner/Operator: ... **UNOLS / US Navy / Woods Hole Oceanographic Institute**  
 Country: ..... **US**  
 Flag: ..... **US**  
 Total number of sister ships already completed: ..... **0**  
 Total number of sister ships still on order: ..... **1**  
 Contract date: ..... **October 2011**  
 Delivery date: ..... **July 2015**

*Neil Armstrong* is the first in class of a new pair of research vessels constructed for the University National Oceanographic Laboratory System (UNOLS) fleet. *Neil Armstrong* and sister ship *Sally Ride* are both auxiliary general oceanographic research ships and will be operated by Woods Hole Institution of Oceanography and Scripps Institution of Oceanography, respectively. Both vessels were designed by GPA in Seattle, Washington, US, and constructed by Dakota Creek Industries (DCI), also of Washington.

The GPA/DCI team strategy for the design of the *Armstrong* class was to meet all critical and optional requirements set forth in the specification developed by the US Navy Office of Naval Research (ONR) and the UNOLS group. While final dimensions were dependent on specification-derived values and space required for desired features, there were also various operational characteristics required in the specification. One of the more critical of these is the prevention bubble sweep-down, which requires a hull form design that discourages flow in way of the bow wake from carrying bubbles generated by the ship's entry downward, toward the instrumentation array located near the mid-ship bottom centerline.

Vibration and noise transmission throughout the hull was another critical operational design element. It was important to prevent mechanically

induced noise on and vibrations from affecting the crew and scientific personnel during their work and resting hours, as well as to prevent adverse effects on onboard sensitive scientific equipment and the marine environment surrounding the vessel. Structural design, including global hull structural analysis and large counterweighted foundations for the main generator sets, along with shock mountings on vibrating equipment and vibration-attenuating coatings, were all employed to ensure that all vibration criteria were met or exceeded. DCI and GPA collaborated with Noise Control Engineering (NCE) to engineer these solutions and validate that these measures were effective. The vessels also ensure minimised effect on the environment by way of both locating all drains and outflows on the port side (opposite the side handling gear and ROV hangar entry), and by the capability to have zero outflow or drainage overboard for up to 72 hours.

*Neil Armstrong* is capable of a sustained speed of 12knots with a range of 11,500nm. The vessel can provide 11.3tonnes of towing force at 4knots and 4.5tonnes at 6 knots in sea state 4 conditions. *Neil Armstrong* and *Sally Ride* are powered by twin AC motors controlled by variable frequency drives provided by Siemens, and propelled by two controllable-pitch Hundested propellers. This arrangement maximises fuel efficiency over a broad operating profile, which includes use of a tunnel stern thruster and a Tees White Gill azimuthing disc bow thruster for station-keeping, while also providing a high degree of control over propeller-induced noise.

For subsea research, the *Armstrong* class is capable of being fitted with a wide array of subsea sensors and sonar equipment on a number of through-hull fittings and array foundations in the transducer room. The transducer room is also equipped with various sea chests and a larger overhead for the movement of scientific equipment to be defined in future years. Flexibility in this space and throughout the design is important in

allowing the *Armstrong* class to adapt to as many different type of missions as possible.

### TECHNICAL PARTICULARS

Length, oa ..... 72.54m  
 Length, bp ..... 70.1m  
 Breadth, moulded ..... 15.24m  
 Depth, moulded ..... 6.7m  
 Gross tonnage ..... 2,602.9tonnes  
 Displacement ..... 3,433.2tonnes  
 Design, draught ..... 4.57m  
 Design, deadweight ..... 1,471.7tonnes  
 Lightweight ..... 1,961tonnes  
 Deck space (total) ..... 174m<sup>2</sup>  
 Service speed ..... 12knots  
 Max speed ..... 12.8knots  
 Bollard pull .... 4.5tonnes @ 4knots @ sea state 4  
 Range (nautical miles) ..... 11,500  
 Daily fuel consumption ..... 11.16tonnes  
 Classification society ..... ABS (under-90m rules)  
 Notations ..... A1, Circle E, AMS, ACCU, NIBS, Ice Class D0, UWILD, 46 CFR Subchapter U, SOLAS (Oceanographic Vessels), MARPOL

Main engine(s)  
 Make ..... Cummins  
 Model ..... QSK38 DM  
 Number ..... 4  
 Output of each engine ..... 1,044kW  
 Propulsion motor(s)  
 Make ..... Siemens  
 Model ..... 1FW-4508-1HA  
 Number ..... 2  
 Output speed ..... 200rpm  
 Propeller(s)  
 Material ..... Stainless steel  
 Manufacturer ..... Hundested  
 Number ..... 2  
 Fixed/controllable pitch ..... Controllable  
 Diameter ..... 2,438mm  
 Speed ..... 200rpm  
 Open or nozzled ..... Open  
 Alternators  
 Make/type ..... Siemens  
 Number ..... 4  
 Output of each set ..... 988kW@60Hz/1,800rpm

Bow thruster(s)  
 Make ..... White Gill 40 T 3S MK V-QR  
 Number ..... 1  
 Output ..... 686kW@677rpm  
 Stern thruster(s)  
 Make ..... Schottel  
 Number ..... 1  
 Output ..... 620kW @ 1,470rpm  
 Deck machinery  
 Crane(s) ..... 1 x Allied TK 70-70, 9.9tonnes@21.3m SWL  
 Winch(es) ..... 2 x Markey hydrographic winches, 10.9tonnes @ 0-62.5m/min  
 1 x Markey traction winch, 11.8tonnes@0-62.5m/min  
 Other deck machinery/equipment ..... 1 x Allied TK4-30 portable crane, 0.9tonnes@9.1m SWL

Bridge electronics  
 Radar(s) ..... JMA- 9100 Series ARPA radar  
 Autopilot ..... Yokogawa / PT500APN2  
 GMDSS ..... JSB -196GM  
 GPS ..... JLR-7500  
 Gyro ..... Navigat X Mk 1  
 Engine monitoring/ fire detection system ..... Salwico CS4000  
 Onboard capacities  
 Fuel oil ..... @98% 556.7m<sup>3</sup>  
 Fresh water ..... @100% 55.4m<sup>3</sup>  
 Sullage ..... @98% 15.7m<sup>3</sup>  
 Ballast water ..... @100% 441.5m<sup>3</sup>  
 Complement .....  
 Crew ..... 20  
 Passengers (scientists) ..... 24  
 Number of cabins ..... 31  
 Other significant or special items of equipment  
 Deep water multibeam survey system  
 Mid-water multibeam survey system  
 Sub-bottom profiler  
 Acoustic Doppler current profiler (ADCP)  
 Acoustic navigation and tracking system





## OCEAN EAGLE 43: Trimaran offshore patrol vessel with monohull-style motions

Builder ..... **CMN Cherbourg**  
 Designer ..... **Nigel Irens Design / CMN Cherbourg**  
 Vessel's name ..... **Ocean Eagle 43**  
 Country ..... **Mozambique**  
 Flag ..... **TBC**  
 Total number of sister ships  
   already completed ..... **2**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **TBC**

The Ocean Eagle 43 (OE43) is a new class of offshore patrol vessel, taking the form of a trimaran or stabilised monohull and incorporating a number of ocean-racing yacht / power trimaran features into its overall design.

The eye-catching vessel's concept matches a slender main hull to floats, for increased stability, range and seakeeping, as well as reduced energy draw and fuel consumption. Stemming from a collaboration between the CMN Cherbourg yard in France, UK naval architect Nigel Irens Design, R&D specialist SeaTeam Aviation and operator ProLarge, the OE43 and her two sisters were developed for operation in Mozambique, where they will primarily undertake fisheries protection and coastal surveillance operations.

The trio was launched in 2015; as *Significant Small Ships of 2015* went to press, the three completed vessels were in Cherbourg, where their future crew members (including the skippers) are undergoing comprehensive training, prior to their imminent delivery.

The OE43 has been described by Nigel Irens Design as a "logical extension" of its previous power trimaran offerings, which include the 34.4m, 1998-built *Cable & Wireless Adventurer*,

the first craft to circumnavigate the world within 74 days. Measuring 43.6m x 15.7m, the OE43 is capable of efficient operation across a range of speeds; whether sprinting or relocating at speeds of up to 30knots, or long-range cruising; in the latter case, the OE43 has recorded a range of more than 3,000nm when travelling at 18knots, and a range of 5,000nm at 12knots.

One of the biggest design challenges concerned the development of the main and float hull forms. Nigel Irens states: "Considerable work was undertaken on the distribution of volume in the hulls both above and below the waterline, in order to find the optimum balance between performance and seakeeping." Equally, the floats received special attention and were tweaked to extract the best desired stability characteristics whilst keeping resistance to a minimum. The end result, Nigel Irens recalls, was a patrol craft "more akin to a monohull than a catamaran".

For additional weight reduction (and subsequent fuel reduction), the OE43 was constructed from lightweight composites – Nigel Irens explains: "The infusion of the main hulls set a new record for a cored epoxy hull infused in one shot." CMN Cherbourg, meanwhile, manufactured the new class of vessels' machinery and systems, to ensure accurate weight control.

To top things off, the OE43 has the capacity to carry, launch and recover a fast-response, 7m-long RIB, featuring a speed of 40knots, and an unmanned aerial vehicle (UAV) with a deck securing system. This configuration enables a wide range of security-related tasks to be undertaken, including anti-piracy / anti-smuggling operations and search and rescue.

### TECHNICAL PARTICULARS

|                                                    |                                                             |
|----------------------------------------------------|-------------------------------------------------------------|
| Length, oa .....                                   | 43.6m                                                       |
| Breadth, moulded .....                             | 15.7m                                                       |
| Depth, moulded .....                               | 3.4m                                                        |
| Gross tonnage .....                                | 140tonnes                                                   |
| Design, draught .....                              | 1.4m                                                        |
| Design, deadweight .....                           | 20tonnes                                                    |
| Deck space (total) .....                           | 37m <sup>2</sup> (aft) / 61m <sup>2</sup> helo (drone) deck |
| Service speed .....                                | 27knots @80% MCR                                            |
| Max speed .....                                    | 30knots                                                     |
| Range (nautical miles) .....                       | 3,000 @ 18knots                                             |
| .....                                              | 5,000 @ 12knots                                             |
| Classification society .....                       | Bureau Veritas                                              |
| Main engine(s)                                     |                                                             |
| Make .....                                         | Scania                                                      |
| Model .....                                        | DI 13 070M                                                  |
| Number .....                                       | 2                                                           |
| Output of each engine .....                        | 404kW                                                       |
| Gearbox(es)                                        |                                                             |
| Make .....                                         | Servogear                                                   |
| Model .....                                        | HD250                                                       |
| Number .....                                       | 2                                                           |
| Output .....                                       | 684rpm                                                      |
| Propeller(s)                                       |                                                             |
| Manufacturer .....                                 | Servogear                                                   |
| Number .....                                       | 2                                                           |
| Fixed/controllable pitch .....                     | Controllable                                                |
| Diameter .....                                     | 1,125mm                                                     |
| Onboard capacities                                 |                                                             |
| Fuel oil .....                                     | 17tonnes                                                    |
| Fresh water .....                                  | 2tonnes                                                     |
| Complement                                         |                                                             |
| Crew .....                                         | 13                                                          |
| Passengers .....                                   | 0                                                           |
| Number of cabins .....                             | 7                                                           |
| Other significant or special items of equipment    |                                                             |
| 7m outboard RIB, mounted on stern ramp             |                                                             |
| Rotary UAV with dedicated deck and securing system |                                                             |





## ORMSÖ: Ice-breaking double-ended car and passenger ferry from Baltic Workboats

Builder ..... **Baltic Workboats**  
 Designer ..... **Baltic Workboats**  
 Vessel's name ..... **Ormsö**  
 Owner/operator ..... **Estonian Maritime Administration**  
 Country ..... **Estonia**  
 Flag ..... **Estonia**  
 Total number of sister ships already completed ..... **0**  
 Total number of sister ships still on order ..... **2**  
 Contract date ..... **March 2013**  
 Delivery date ..... **May 2015**

Delivered to the Estonian Maritime Administration in May 2015, the 45m *Ormsö* is the first in Baltic Workboats' resilient new breed of Arc4 ice-classed, double-ended car and passenger ferries. Constructed in accordance with the Russian Maritime Register of Shipping (RS), and to comply with the latest EU directives, the vessel is manned by a crew of six and has the capacity to carry up to 200 passengers and 32 cars or two trucks.

The awarding of Arc4 ice class – which necessitates a similar set of requirements to Finnish/Swedish 1A ice class – reflects the cold environments in which *Ormsö* and her two sisters (*Kihnu Virve*, delivered in October 2015, and *Soela*, scheduled for delivery in 2017) will be operating. The ferry is powered by a pair of MTU 8V4000 engines, each producing 1,000kW of power and feeding Rolls-Royce azimuth thrusters. Two Volvo Penta generators have also been installed, to provide additional onboard power. Together, this arrangement grants the vessel an operating speed of 15knots when operating in calm water, and 3knots when ploughing through 450mm of solid ice.

*Ormsö* and her sisters have also been developed with adaptability in mind. For instance, the design enables the owner to install a twin deck, if needs be, thus creating a second car level that can accommodate an extra 10 cars.

### TECHNICAL PARTICULARS

Length, oa ..... 45m  
 Length, bp ..... 44.07m  
 Breadth, moulded ..... 12m  
 Depth, moulded ..... 4.7m  
 Gross tonnage ..... 926tonnes  
 Displacement ..... 690tonnes  
 Design, draught ..... 2.7m  
 Design, deadweight ..... 127tonnes  
 Lightweight ..... 550tonnes  
 Deck space (total) ..... 400m<sup>2</sup>  
 Deck capacity ..... 90tonnes/400m<sup>2</sup>  
 Service speed ..... 12knots  
 Max speed ..... 13.5knots  
 Range (nautical miles) ..... 3,000 (@ economical speed and full bunkers = 40m<sup>3</sup>)  
 Classification society ..... Russian Maritime Register of Shipping  
 Notations ..... KM<sup>0</sup>Arc4(1) R3  
 AUT Ro-Ro Passenger Ship  
 Main engine(s)  
   Make ..... MTU  
   Model ..... 8V 4000 M63  
   Number ..... 2  
   Output of each engine ..... 1,000kW  
 Thruster(s)  
   Make ..... Rolls-Royce  
   Model ..... US 155 P14 FP  
   Number ..... 2  
 Propeller(s)  
   Material ..... Bronze  
   Manufacturer ..... Rolls-Royce  
   Number ..... 2

Fixed / controllable pitch ..... Fixed  
 Diameter ..... 1,600mm  
 Open or nozzled ..... Open  
 Alternators  
   Make/type ..... Volvo Penta  
   Number ..... 2  
   Output of each set ..... 175kW @ 1,500rpm  
 Deck equipment ..... 2 x Data Hidrolik DHC 3000 H mooring capstans, max pull 3.5tonnes  
   1 x Data Hidrolik DZC 7000H windlass, max pull 7.7tonnes  
   1 x Davit International D-CRM. FR.21/4,5 davit for fast rescue boat, SWL 16kN  
 Bridge electronics  
   Radar(s) ..... Simrad Argus 12U/6X  
   Autopilot ..... Simrad AP80  
   GPS ..... Simrad MX610 DGPS  
   Gyro ..... Simrad CS85  
   Chart plotter ..... Simrad CS68 PLECDIS  
   Engine monitoring system ..... ComAp  
   Fire detection system ..... Deckma  
 Onboard capacities  
   Fuel oil ..... 40m<sup>3</sup>  
   Fresh water ..... 12m<sup>3</sup>  
   Sullage ..... 12m<sup>3</sup>  
   Ballast water ..... 150m<sup>3</sup>  
 Complement  
   Crew ..... 6  
   Passengers ..... 200  
   Number of cabins ..... 5  
 Vehicles  
   Number of vehicle decks ..... 1 (with option for additional platform deck for 10 cars)  
   Number of cars ..... 32 personal cars  
   Number of trucks/trailers ..... 2 x 45tonne / 16.5m-long trucks, or 4 x 20tonne trailers



## OSSOY: First medium-sized AHTS class constructed by Sinopacific, China

Builder ..... **Sinopacific Shipbuilding Group**  
 Designer ..... **Shanghai Design Associates (SDA)**  
 Vessel's name ..... **OSSOY**  
 Owner/operator ..... **Femco Group**  
 Country ..... **Russia**  
 Flag ..... **Liberia**  
 Total number of sister ships  
   already completed ..... **2**  
 Total number of sister ships still on order ..... **2**  
 Contract date ..... **February 2013**  
 Delivery date ..... **April 2015**

**O**SSOY is the first vessel in Sinopacific Shipbuilding's SPA150 class, and constitutes the first medium-sized anchor handling tug supply (AHTS) vessel in the builder's in-house SP series. Featuring just under 9,000kW of propulsive power, a Class 2 dynamic positioning system (DP2) and a 150tonne bollard pull, the vessel is suitable for a variety of offshore tasks, including tug work, anchoring, external fire-fighting, oil spill response and recovery and loading various liquids and dry cargoes.

The vessels in the SPA series have been rolled out for the rapidly expanding owner and operator Far East Marine Company (Femco), which, since its establishment in 1998, has come to oversee a current fleet of 11 AHTS vessels, two dry cargo vessels and one fast support craft. Sinopacific comments: "The design team has been able to offer flexibility and specific configurations for each vessel, so as to customise each one to Femco's specific demands."

Sinopacific collaborated closely with Rolls-Royce Marine, from the earliest stage of the design process, to create an optimised propulsive solution for the SPA150 series. Reportedly, this is the first time that Rolls-Royce has provided integrated equipment packages for vessels on build in China that were not of that company's own design.

OSSOY is arranged as a single-deck vessel, with machinery and accommodation units (including

36 cabins) located forward. Other design features include a bulbous bow, intended to significantly reduce wave resistance when the 72m x 17.2m vessel is sailing in transit, thereby providing a bulwark against excessive fuel consumption and strengthening the vessel's 'green' credentials.

### TECHNICAL PARTICULARS

Length, oa ..... 72m  
 Length, bp ..... 62.88m  
 Breadth, moulded ..... 17.2m  
 Depth, moulded ..... 7.8m  
 Gross tonnage ..... 2,721tonnes  
 Displacement ..... 5,376.1tonnes  
 Design, draught ..... 5m  
 Design, deadweight ..... 1,506tonnes  
 Lightweight ..... 2,506tonnes  
 Deck space (total) ..... 516m<sup>2</sup>  
 Deck capacity ..... 10tonnes/m<sup>2</sup>  
 Service speed ..... 15.65knots @ 85% MCR  
 Max speed ..... 16.3knots  
 Bollard pull ..... 150tonnes  
 Range (nautical miles) ..... 6,250  
 Daily fuel consumption ..... 35.6tonnes  
 Classification society ..... Bureau Veritas  
 Notations ..... \*HULL, \*MACH  
   Unrestricted Navigation Tug/Supply Vessel  
   Oil Product and LHNS/fire-fighting  
   ship 1 Water spray / special  
   service – anchor handling  
   \*AUT-UMS, \*DYNAPOS  
   AM/ AT/ R / CLEANSHIP /  
   PROTECTED FO TANKS

Main engine(s)  
 Make ..... Rolls-Royce Marine  
 Model ..... Bergen B32:40L9PCD  
 Number ..... 2  
 Output of each engine ..... 4,500kW  
 Gearbox(es)  
 Make ..... Rolls-Royce Marine  
 Model ..... 850 AGHC-S-B-P60  
 Number ..... 2  
 Output ..... 138rpm

Propeller(s)  
 Material ..... NiAlBr  
 Manufacturer ..... Rolls-Royce Marine  
 Number ..... 2  
 Fixed/controllable pitch ..... Controllable  
 Diameter ..... 4,000mm  
 Speed ..... 138rpm  
 Open or nozzled ..... Nozzled  
 Alternators  
 Make/type ..... Marelli MJB500 SC4  
 Number ..... 2  
 Output of each set ..... 1,500kW  
 Bow thruster(s)  
 Make ..... Rolls-Royce Marine  
 Number ..... 2  
 Output of each ..... 690kW @ 1,180rpm  
 Stern thruster(s)  
 Make ..... Rolls-Royce Marine  
 Number ..... 1  
 Output of each ..... 690kW @ 1,180rpm  
 Deck machinery ..... 1 x crane  
   (Chinese manufacturer, unspecified),  
   2tonnes SWL  
   1 x Rolls-Royce  
   Marine winch, 150tonnes  
   1 x roller  
   (Chinese manufacturer, unspecified)  
 Bridge electronics  
 Radar(s) ..... Furuno  
 Autopilot ..... Furuno  
 GMDSS ..... Furuno  
 GPS ..... Furuno  
 Gyro ..... Rolls-Royce  
 Chart plotter ..... Furuno  
 Engine monitoring/  
   fire detection system ..... Rolls-Royce/Consilium  
 Onboard capacities  
 Fuel oil ..... 875m<sup>3</sup>  
 Fresh water ..... 600m<sup>3</sup>  
 Sullage ..... 38m<sup>3</sup>  
 Ballast water ..... 1,190m<sup>3</sup>  
 Complement  
 Crew ..... 16  
 Passengers ..... 20  
 Number of cabins ..... 36





## PELICAN PRIDE: Economic offshore FSIV with additional creature comforts

Builder ..... **Penguin Shipyard International**  
 Designer ..... **Penguin Shipyard International**  
 Vessel's name ..... ***Pelican Pride***  
 Owner/operator ..... **Pelican Offshore Services**  
 Country ..... **Singapore**  
 Flag ..... **Malaysia**  
 Total number of sister ships  
   already completed ..... **2**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **Q1 2015**  
 Delivery date ..... **Q2 2015**

Mid-sized crewboat specialist Penguin Shipyard International rolled out the latest innovation in its line of 'Flex' vessels in Q2 2015, with the delivery of two of its new Flex-45 class fast supply intervention vessels (FSIVs), including the debutante *Pelican Pride*, to Pelican Offshore, the shipowning subsidiary of Penguin International Limited.

Designed by Penguin Shipyard, the Flex-45 Executive FSIV is powered by four mechanically controlled Baudouin M26.2 main engines with a combined output of 3,580kW. Performance-wise, the Flex-45 is not designed to be a high-end, high-speed, high-cost FSIV. Instead, it is designed to transport personnel and cargo economically and efficiently, while offering "executive levels" of comfort, Penguin claims. The overall effect, the designer and builder adds, is one of "style meets sensibility".

The vessel has a respectable top speed of 27knots with 45dwt and a corresponding fuel consumption of just 840litres per hour. At economical speed, the fuel consumption drops to a modest 680litres per hour. The FSIV's 135m<sup>2</sup> clear deck area can transport up to 80tonnes of cargo and includes a Palfinger knuckle-boom crane as standard equipment.

Despite taking an economical approach, Penguin Shipyard has not skimped on creature comforts. For example, each Flex-45 comes with 60 reclining business class seats as a standard feature, plus four crew cabins with a total of 10 bunks.

And, should the Flex-45 be required to undertake firefighting duties, the Flex-45 is equipped with

two FiFi monitors capable of throwing out 1200m<sup>3</sup> of seawater per hour.

"The Flex-45 is designed to be economical to own and operate, and comfortable to ride and sleep in," James Tham, managing director of Penguin International Limited, parent company of Penguin Shipyard, states. "This is an FSIV that makes good economic sense, considering the current state of the offshore industry."

### TECHNICAL PARTICULARS

Length, oa ..... 45m  
 Length, bp ..... 40m  
 Breadth, moulded ..... 8.8m  
 Depth, moulded ..... 3.65m  
 Gross tonnage ..... 352tonnes  
 Displacement ..... 323tonnes  
 Design, draught ..... 1.9m  
 Design, deadweight ..... 170tonnes  
 Lightweight ..... 150tonnes  
 Deck space (total) ..... 135m<sup>2</sup>  
 Deck capacity ..... 2tonnes/m<sup>2</sup>  
 Service speed ..... 25knots @ 85% MCR /45m dwt  
 Max speed ..... 27knots  
 Range (nautical miles) ..... 2,000-3,000  
 Fuel consumption ..... 680litres/hr @service speed  
 Classification society ..... Bureau Veritas  
 Notations ..... \*Hull \*Mach, Crew Boat  
 Other important international regulations  
 complied with ..... MARPOL, SOLAS, IMDG  
 Main engine(s)  
   Make ..... Baudouin  
   Model ..... 12M26.2 (IMO Tier 2)  
   Number ..... 4  
   Output of each engine ..... 895kW @ 1,950rpm  
 Gearbox(es)  
   Make ..... Twin Disc  
   Model ..... MGX-6690SC  
   Number ..... 4  
   Output ..... Ratio 2.93:1  
 Propeller(s)  
   Material ..... NiAlBr  
   Manufacturer ..... Twin Disc / Rolla

Number ..... 4  
 Fixed / controllable pitch ..... Fixed  
 Diameter ..... 1.245m  
 Speed ..... 665.5rpm (@ max torque)  
 Open or nozzled ..... Open  
 Alternators  
   Make/type ..... Leroy-Somer LSAM 44.2 L12  
   Number ..... 3  
   Output of each set ..... 92kW  
 Bow thruster(s)  
   Make ..... Global Marine Engineering  
   Number ..... 1  
   Output of each ..... 98.4kW @ max 1,100rpm  
 Deck machinery ..... 1 x Palfinger PK  
   4501 knuckle-boom crane, 0.5tonnes@6m  
   1 x Hypac HHA W2520 anchor windlass  
 Bridge electronics  
   Radar(s) ..... Simrad 12U/6X and NSS 12"  
   Autopilot ..... Simrad AP70  
   GMDSS ..... Sailor 6222  
   Other communications systems ..... VHF Radio  
   Sailor 6222  
   MF/HF Radio Sailor 6301  
 GPS ..... Simrad GN70  
 Gyro ..... Simrad RGC80  
 Fire detection system ..... Tyco Fire, Security  
   & Service  
 Onboard capacities  
   Fuel oil ..... 135,000litres  
   Fresh water ..... 35,000litres  
   Sullage ..... 600litres  
 Complement  
   Crew ..... 10  
   Passengers ..... 60  
   Number of cabins ..... 4 (below main deck; 1 x  
     4-man, 3 x 2-man)  
     2 (VIP, on main deck; 2 x 2-man)  
 Other significant or special items of equipment  
   2 x external firefighting monitors with remote  
     clutch and controls, 2 x 600m<sup>3</sup>/hr  
   Motorised workboat



## PRINCE JOB I: New OSV for African offshore fields, with a multipurpose remit

Builder ..... **Shipyards De Hoop**  
 Designer ..... **Shipyards De Hoop**  
 Vessel's name ..... **Prince Job I**  
 Owner/operator ..... **Awaritse**  
 Country ..... **Nigeria**  
 Flag ..... **Nigeria**  
 Total number of sister ships  
 already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **January 2014**  
 Delivery date ..... **March 2015**

The offshore market may have remained stagnant in 2015, but innovative solutions continue to abound, as is evident in the delivery of the Shipyards De Hoop-designed and -built multipurpose OSV *Prince Job I*. Developed to operate in Chevron's offshore oil fields near Nigeria, the 68.5m vessel's duties include the discharge of transmix liquids; although such work is usually handled by tankers, draught restrictions affecting the operational zone have made these latter vessel types unsuitable. Similarly, tankers would prove poor platforms for *Prince Job I*'s other stipulated tasks, such as transportation of deck cargo and inspection, repair and maintenance (IRM) and firefighting operations. IRM work is especially important, given that the transmix liquids are being pumped from the offshore site to shore via a single Chevron pipeline; this pipeline must be kept in optimal working order, and repaired, should any defects occur, in order to reduce the likelihood of accidents, which could include life-threatening incidents such as explosions.

The vessel's cargo area spans some 500m<sup>2</sup> and has been strengthened to 5tonnes per m<sup>2</sup>, with container fittings and lashing points enabling her to accommodate a containerised cargo equivalent to 16teu. Two Sormec offshore knuckle boom

cranes, each rated 10tonnes at an outreach of 15m, have been installed to assist hose-handling and loading/unloading operations. For maximum, unobstructed access across the cargo deck, the superstructure has been situated as far forward as possible, De Hoop explains.

*Prince Job I* has been fitted with four inflatable life rafts, split equally between the port and starboard sides, each featuring the capacity for 25 persons, and an aluminium workboat. Should the vessel be called on to tackle fires, a pair of electric pumps, each with a capacity of 1.5 million litres, can feed two monitors with a combined capacity of 2.4 million litres per hour. This set-up has earned the vessel DNV-GL FiFi 1 status.

The superstructure consists of 4.5 levels and contains the wheelhouse (with the bridge split into a forward-facing section, for vessel transit, and an aft-facing section, where DP-relevant operations are conducted) and 10 cabins, providing accommodation for up to 30 crew members.

### TECHNICAL PARTICULARS

Length, oa ..... 68.45m  
 Length, bp ..... 62.58m  
 Breadth, moulded ..... 15.77m  
 Depth, moulded ..... 5.5m  
 Gross tonnage ..... 1,847tonnes  
 Displacement ..... 3,863tonnes  
 Design, draught ..... 4.62m  
 Design, deadweight ..... 2,300tonnes  
 Lightweight ..... 1,465tonnes  
 Deck space (total) ..... 500m<sup>2</sup>  
 Deck capacity ..... 5tonnes/m<sup>2</sup>  
 Service speed ..... 11knots  
 Max speed ..... 12knots  
 Range (nautical miles) ..... 7,200  
 Daily fuel consumption ..... 6tonnes  
 Classification society ..... Germanischer Lloyd  
 Notations ..... ✱100A5 Offshore Service Vessel  
 ✱MC, DP2, ✱FI-FI-1, EP-D, OR, HLNS  
 Bollard pull ..... 30tonnes

Main engine(s)  
 Make ..... Caterpillar  
 Model ..... C32 (x2), C18 (x2)  
 Number ..... 4  
 Output of each engine ..... 950kW (C32) / 550kW (C18)

Propeller(s)  
 Material ..... NiAlBr  
 Manufacturer ..... Veth (azimuth)  
 Number ..... 2  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1,900mm  
 Open or nozzled ..... Nozzled

Bow thruster(s)  
 Make ..... Veth  
 Number ..... 2  
 Output of each ..... 450kW / 422rpm

Deck machinery  
 2 x Sormec cranes, 10tonnes @15m SWL each  
 2 x C-Nautical tugger winches, 10tonnes each  
 1 x Kraaijeveld roller, 100tonnes

Bridge electronics  
 Radar(s) ..... 2 x JRC JMA-5312-6  
 Autopilot ..... NAVIS (integrated in DP)  
 GMDSS ..... JRC  
 GPS ..... JRC JLR-7800  
 Gyro ..... 3 x Alphatron Minicourse  
 Chart plotter ..... Navtex JRC NCR-333  
 Engine monitoring/ fire detection system ..... MiniMax

Onboard capacities  
 Fuel oil ..... 520m<sup>3</sup>  
 Fresh water ..... 240m<sup>3</sup>  
 Ballast water ..... 1,000m<sup>3</sup>  
 Liquid mud ..... 340m<sup>3</sup>  
 Brine ..... 300m<sup>3</sup>  
 Transmix fluid ..... 800m<sup>3</sup>

Complement  
 Crew ..... 30  
 Passengers ..... 0  
 Number of cabins ..... 10











## TAI PARI: First of two customised China-built tugs, complete with optimised Z-drive units

Builder ..... **Hin Lee (Zhuhai) Shipyard Co, Ltd (Cheoy Lee)**  
 Designer ..... **Robert Allan Ltd**  
 Vessel's name ..... **Tai Pari**  
 Owner/operator ..... **The Port of Tauranga Limited**  
 Country ..... **New Zealand**  
 Flag ..... **New Zealand**  
 Total number of sister ships already completed ..... **1**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **December 2013**  
 Delivery date ..... **June 2015**

As part of its fleet renovation programme, the Port of Tauranga Limited, New Zealand, took delivery of a pair of tug newbuilds, constructed in China to a customised version of Robert Allan Ltd's (RAL's) Ramparts 2400W class design. *Tai Pari* was delivered in June 2015 (her sole sister, *Tai Timu*, followed hot on her heels later on in the month), with design input based on the Port of Tauranga's operational specifications. The contract was certainly a significant one for Cheoy Lee, being its first RAL-designed construction project to date.

The now familiar Ramparts hull design incorporates an enhanced dead rise for superior seakeeping, as well as RAL's conically shaped double chine stern for effective control and vessel stability at high speed. The Port additionally requested a vessel type that would be capable of towing, albeit with a high side stepping speed. As a result, RAL conducted a detailed CFD study to determine the optimal skeg geometry to yield the desired effects. Tests were conducted, utilising modelled Z-drive thrusters and rotating propellers. Another challenge was to restrict the hull length, so that it would fall under the 24m rule length for load line and tonnage conventions.

With environmental protection in mind, *Tai Pari*'s main Caterpillar 3516C engines, each rated 2,350kW at 1,800rpm, are 'C' rated, compliant with IMO Tier II guidelines. *Tai Pari* has also been fitted with firefighting equipment, including a single centrifugal pump, which is fed by the port main engine, and a forward-mounted monitor, the latter featuring a dispersal capacity of 1,200m<sup>3</sup> per hour.

*Tai Pari* and *Tai Timu* can each accommodate up to six members of crew. Each tug's forward lower deck houses a pair of two-man crew rooms, complete with laundry and washroom facilities; meanwhile, the master and chief engineer enjoy their own individual cabins on the main deck.

### TECHNICAL PARTICULARS

Length, oa ..... 24.4m  
 Breadth, moulded ..... 11.25m  
 Depth, moulded ..... 4.2m  
 Gross tonnage ..... 310.4tonnes  
 Displacement ..... 517tonnes  
 Design, draught ..... 5.34m  
 Design, deadweight ..... 105tonnes  
 Lightweight ..... 412tonnes  
 Service speed ..... 10knots (25% MCR)/  
 12knots (50% MCR)  
 Max speed ..... 13knots  
 Bollard pull ..... 72tonnes  
 Range (nautical miles) ..... 2,040 (at 10knots)  
 Daily fuel consumption ..... 7.5tonnes  
 Classification society ..... Lloyd's Register  
 Notations ..... \*100A1 Tug,  
 \*IWS, \*LMC, UMC  
 for restricted service

Main engine(s)  
 Make ..... Caterpillar  
 Model ..... 3516C HD  
 Number ..... 2

Output of each engine ..... 2,350kW @ 1,800 rpm  
 Z-drive(s)  
 Make ..... Rolls-Royce  
 Model ..... US 255 FP  
 Number ..... 2  
 Output ..... 240rpm  
 Material ..... NiAlBr  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 2,600mm  
 Special adaptations ..... TK  
 (high performance) nozzle  
 Open or nozzled ..... Nozzled

Alternator(s)  
 Make ..... Caterpillar C6.6  
 Number ..... 2  
 Output of each set ..... 125kWe @ 1,500rpm  
 Deck machinery  
 1 x MacGregor winch, 175tonnes  
 brake capacity  
 1 x MacGregor capstan,  
 5tonnes@0-15m/min  
 1 x Mampaey tow hook, 100tonnes SWL  
 Bridge electronics  
 GMDSS ..... Furuno A3  
 AIS ..... Garmin AIS 300  
 Chart plotter ..... Garmin  
 Engine monitoring/  
 fire detection system ..... Wilhelmsen  
 Onboard capacities  
 Fuel oil ..... 76.8m<sup>3</sup>  
 Fresh water ..... 10.1m<sup>3</sup>  
 Grey water ..... 2.8m<sup>3</sup>  
 Black water ..... 2.8m<sup>3</sup>  
 Ballast water ..... 45.8m<sup>3</sup>  
 Complement  
 Crew ..... 6  
 Passengers ..... 0  
 Number of cabins ..... 4







## TRIDENT: Versatile research vessel for academic projects and training

Builder ..... **Geo Shipyard, Inc**  
 Designer ..... **Roger Fyffe**  
 Vessel's name ..... **Trident**  
 Owner / operator ..... **Texas A&M University**  
 Country ..... **US**  
 Flag ..... **US**  
 Total number of sister ships  
   already completed ..... **0**  
 Total number of sister ships still on order ..... **0**  
 Contract date ..... **May 2014**  
 Delivery date ..... **May 2015**

Louisiana is one of the US' most thriving boatbuilding hubs at present, and New Iberia-based Geo Shipyard added to the state's tally in 2015 with the launch and delivery of the 19.8m loa aluminium catamaran research vessel *Trident*. Like a fellow entrant in this year's *Significant Small Ships* (see *London Titan*, pages 42-43), *Trident* has been described as a 'Swiss Army Knife' in lieu of her versatile nature, enabling a number of research-related and academic studies to be conducted on board, as well as providing a platform for training.

Reported to be the first new vessel that the University of Texas, based in Galveston, has ordered in 40 years, *Trident* has been developed to operate 24/7 for up to five days, between port calls, and to travel up to 200nm from shore. The vessel has been fitted with a 22.3m<sup>2</sup> dry lab, replete with computer facilities, and boasts a spacious aft deck capable of holding a wet lab counter, comprising a deep sink with fresh /salt water taps and a fresh water shower. *Trident's* rear cargo deck measures some 34.8m<sup>2</sup>.

Deck equipment includes: a DT Marine-manufactured DT303EM survey/trawl winch and an aluminium A-frame; a 2tonne-capacity knuckle boom crane with a maximum reach of 6.86m; a fire pump, supported by two fire stations; and a 4.6m inflatable RIB. The inclusion of two Wesmar V2 thrusters is intended to

enhance *Trident's* manoeuvrability, especially in areas where space is restricted. Kobelt supplied the vessel's steering system.

Inside, three staterooms accommodate 12 beds, and scientist passengers / students / trainees can also take advantage of the pilothouse, a galley and a mess dining area. The main Scania 13litre diesel engines, rated 373kW apiece, connect to 812.8mm x 1,067mm ZF four-bladed nibrals wheels through ZF 360 marine gears with 2:48:1 reduction ratios. Together with the vessel's twin Wesmar V2 thrusters, this propulsion package grants *Trident* a cruising speed of 17knots, increasing to 23knots max.

### TECHNICAL PARTICULARS

Length, oa ..... 19.8m  
 Length, bp ..... 19.1m  
 Breadth, moulded ..... 6.55m  
 Depth, moulded ..... 2.84m  
 Gross tonnage ..... 92tonnes  
 Displacement ..... 48tonnes  
 Design, draught ..... 1.37m  
 Design, deadweight ..... 10.38tonnes  
 Lightweight ..... 37.72tonnes  
 Deck space (total) ..... 54.9m<sup>2</sup>  
 Deck capacity ..... 4tonnes/m  
 Service speed ..... 17knots @ 75% MCR  
 Max speed ..... 23knots  
 Range (nautical miles) ..... 1,200  
 Daily fuel consumption ..... 2,271litres  
 Classification society ..... US Coast Guard-inspected

Main engine(s)  
   Make ..... Scania  
   Model ..... DI 13 070  
   Number ..... 2  
   Output of each engine ..... 373kW  
 Gearbox(es)  
   Make ..... ZF Marine  
   Model ..... ZF 360

Number ..... 2  
 Output ..... 725rpm  
 Propeller(s)  
   Make ..... NiAlBr  
   Manufacturer ..... ZF Marine  
   Number ..... 2  
   Fixed / controllible pitch ..... Fixed (41.5)  
   Diameter ..... 810mm  
   Speed ..... 725rpm  
   Special adaptations ..... Anti-singing edge  
   Open or nozzled ..... Open  
 Alternators  
   Make ..... Westerbeke  
   Number ..... 2  
   Output of each set ..... 33kW  
 Bow thruster(s)  
   Make ..... Wesmar  
   Number ..... 2  
   Output of each ..... 17.15kW  
 Deck machinery  
   1 x EST crane, 2tonnes SWL  
   1 x DT Marine winch, 0.34tonnes capacity  
 Bridge electronics  
   Radar(s) ..... 2 x Furuno MD12  
   Autopilot ..... Simrad AP28  
   Other communications systems ..... Furuno Hailer  
   GPS ..... Furuno / Rosepoint  
   Gyro ..... Simrad  
   Chart plotter ..... Furuno/Rosepoint  
   Fire detection system ..... Fire Boy  
 Onboard capacities  
   Fuel oil ..... 9,464litres  
   Fresh water ..... 1,268litres  
 Complement  
   Crew ..... 4  
   Passengers ..... 48  
   Number of cabins ..... 3  
 Other significant or special items of equipment  
   Large dry lab facility  
   SK 450 GPD water maker



## UMOE VENTUS: Wind farm support craft combining hovercraft and cat technology

Builder..... **Umoe Mandal**  
 Designer..... **Umoe Mandal**  
 Vessel's name..... **Umoe Ventus**  
 Owner/operator..... **Umoe Wind**  
 Country..... **Norway**  
 Flag..... **Denmark**  
 Total number of sister ships  
 already completed..... **1**  
 Total number of sister ships still on order..... **1**  
 Contract date..... **October 2013**  
 Delivery date..... **March 2015**

The launch and delivery of the 27.4m x 10m offshore wind park crew transfer vessel (CTV) *Umoe Ventus* ticked off a number of 'firsts', as: the first-in-class in a new series of 'Wave Craft' high-speed CTVs; the first renewable energy support vessel to be supported by the Carbon Trust; and the first of these vessel types to be put together by Norwegian designer and boatbuilder Umoe Mandal.

The genesis of *Umoe Ventus* can be traced back to 2010, when the Carbon Trust launched a ship design competition to address the key challenge of how best to enable wind turbine technicians to safely access /operate on farms located in deeper waters. The Carbon Trust's Offshore Wind Accelerator (OWA) programme, set up in 2008, has worked alongside a number of energy companies – including DONG Energy, Statoil, Scottish Power and E.ON – to develop new solutions for the offshore wind farm support sector.

The winner of the ship design competition, selected from 450 entries, was the Wave Craft class proposed by Umoe Mandal, itself a newcomer to this sector; previously, the designer / builder had specialised in producing high-speed naval corvettes.

In fact, *Umoe Ventus* ' design borrows liberally from the 'surface effect ship' (SES) concept, one that has proven popular in the Norwegian Navy, and which has been credited as one of the fastest vessel types available; speeds of 60knots and more are not uncommon. The SES concept incorporates a combination of catamaran hulls and hovercraft

technology, deploying an air cushion between the hulls to lift 80% of the vessel above water. As a result, the SES is able to curb wave-induced motions and attain higher speeds than the average catamaran whilst burning less fuel.

In addition, when the air cushion is raised, the vessel can realise a minimum draught of approximately 0.7m, thereby enabling her to access ports subject to tidal restrictions and shallow waters.

Reduced wave-induced motions equate to greater crew comfort. "Seasickness is almost non-existent," comments Umoe Mandal's Trygve Halvorsen Espeland, development manager for the Wave Craft. "Further, the air cushion damps motions when accessing turbines, increasing the operational window to wave heights of up to 2.5m significant wave height." The Carbon Trust has calculated that the Wave Craft design is able to increase wind farm accessibility by "at least 20%".

Her owner, Umoe Wind (Umoe Mandal's sister company), has handed *Umoe Ventus* over to DONG Energy, which has chartered the vessel to support its activities at the 312MW Borkum Riffgrund 1 offshore wind farm, in the North Sea.

### TECHNICAL PARTICULARS

Length, oa..... 27.2m  
 Length, bp..... 23.9m  
 Breadth, moulded..... 10.2m  
 Depth, moulded..... 0.7m (on cushion)  
 3m (off cushion)  
 Gross tonnage..... 217tonnes  
 Displacement..... 103tonnes  
 Design, draught..... 0.7m  
 Design, deadweight..... 14.9tonnes  
 Lightweight..... 88tonnes  
 Deck space (total)..... 50m<sup>2</sup>  
 Deck capacity..... 0.65tonnes / 1tonne in  
 load-carrying areas  
 Service speed..... 40knots @100% MCR  
 Max speed..... 42knots  
 Range (nautical miles)..... 700  
 Daily fuel consumption..... 2.9tonnes  
 Bollard pull..... 15tonnes  
 Classification society..... DNV GL

Notations..... \*1A1 HSLC R1  
 Wind Farm Service 1  
 Other important international  
 regulations complied with..... Danish Maritime  
 Authority notice B  
 Main engine(s)  
 Make..... MTU  
 Model..... 2000 16V M72  
 Number..... 2  
 Output of each engine..... 1,440kW  
 Gearbox(es)  
 Make..... Transtech (2)  
 WCG-Bevel (2)  
 Number..... 4  
 Output speed..... 970rpm  
 Alternators  
 Make / type..... Perkins – Goodlike JDS284  
 Number..... 2  
 Output/speed of each set..... 64kW / 1,500rpm  
 Waterjet(s)  
 Make..... MJP 650 CSV  
 Number..... 2  
 Deck machinery  
 Crane(s)..... 1 x 1.5tonnes/m SWL (optional)  
 Winch(es)..... 1 x MBH anchor winch  
 Other deck equipment..... 1 x 1tonne capstan  
 Bridge electronics  
 Radar(s)..... 2 x Furuno Marine ARPA  
 Autopilot..... Simrad AP70  
 GMDSS..... Sailor VHF6222 /  
 6215 MF/HF N163  
 GPS..... Furuno GP-150/017S  
 GPS compass..... Furuno AP 70  
 Chart plotter..... TECDIS  
 Engine monitoring system..... MCS Evolution V5  
 Fire detection system..... Consilium M4.3  
 Onboard capacities  
 Fuel oil..... 15tonnes  
 Fresh water..... 1.5tonnes  
 Sullage..... 0.7tonnes  
 Ballast water..... 6tonnes  
 Complement  
 Crew..... 2-3  
 Passengers..... 12  
 Number of cabins..... 2





## YENIÇAY 1: 'Big sister' to RAL's previous RAscal 1500 class makes her debut

Builder ..... **Sanmar Shipyard**  
 Designer ..... **Robert Allan Ltd.**  
 Vessel's name ..... **Yeniçay 1**  
 Owner/operator ..... **Sanmar**  
 Country ..... **Turkey**  
 Flag ..... **Turkey**  
 Total number of sister ships  
 already completed ..... **1**  
 Total number of sister ships still on order ..... **2**  
 Contract date ..... **Unspecified**  
 Delivery date ..... **January 2015**

Robert Allan Ltd (RAL) continued its long-standing relationship with Turkey's Sanmar Shipyard in 2015, one of the notable deliveries being the first of RAL's new RAscal 1800 class of azimuth stern drive (ASD) tug.

Incorporated directly into Sanmar's fleet, *Yeniçay 1* and the RAscal 1800 series builds on RAL's previous RAscal 1500 class, and has been described as the 'big sister' to that particular breed of twin Z-drive tug. Whereas the RAscal 1500 measured some 14.9m x 8.3m – as featured on the vessel *Brage*, which Sanmar delivered to Norwegian owner Bukser og Berging in 2014 – the RAscal 1800 features an extended length of 18.7m loa and 9.2m, with a corresponding increase in vessel depth.

*Yeniçay 1*'s propulsive set-up comprises: two Caterpillar C32 diesels, each rated 970kW at 1,800rpm; a pair of Veth VZ-900 fixed pitch Z-drive units; and five-bladed, nozzled, 1,700mm-diameter propellers. This set-up grants the tug a bollard pull of 31tonnes and a free running speed of 12knots. Designed for a four-man crew, *Yeniçay 1* features two below-deck double cabins, a galley, a mess and toilet/shower facilities. By incorporating resiliently mounted engines, visco-elastic floor technology and

composite shafting into her design, onboard noise levels have been restricted to 63dbA in the wheelhouse and 60dbA in the galley area and cabins.

The vessel's primary tasks include ship-assist work, line-handling, inland coastal towing and general harbour duties. Additionally, she has also been equipped for firefighting operations. In addition to her spacious fuel and water tanks, the tug can store up to 1,618litres of foam, and the vessel has been fitted with a fi-fi pump, courtesy of Fire Fighting Systems (FFS), with a capacity of 600m<sup>3</sup> per hour.

The bow and stern are equipped with stainless steel fairleads, to accommodate a towline during her ship-assist operations, and the bow also features a DMT TW ship assist hawser winch, which is rated 0-30m/minute at 3tonnes. The hawser drum contains 100m worth of 40mm nylon rope, arranged in four layers, with a brake capacity of 75tonnes. RAL explains: "The hawser winch forward is used for normal ship-handling arrangements but the towline can also be passed aft through a tunnel beneath the wheelhouse, and then through a towing portal on the aft deck, for aft towing operations." A 30tonne SWL towing hook is positioned aft, for ship-handling tows.

Bigger may be yet to come; Sanmar is reportedly working on a 'super-powered' version of the RAscal 1800, which would replace the 1800's C32 engines with Caterpillar 3512 units, which would boost the bollard pull to 40tonnes.

### TECHNICAL PARTICULARS

Length, oa ..... 18.7m  
 Length, bp ..... 17.4m  
 Breadth, moulded ..... 9.2m  
 Depth, moulded ..... 3.5m  
 Gross tonnage ..... 146tonnes  
 Displacement ..... 256tonnes

Design, draught ..... 3.7m  
 Design, deadweight ..... 44tonnes  
 Lightweight ..... 212tonnes  
 Service speed ..... 12knots  
 Max speed ..... 12knots  
 Range (nautical miles) ..... 800  
 Daily fuel consumption ..... 9.7tonnes  
 Classification society ..... ABS  
 Notations ..... ✱A1, Towing Vessel,  
 ✱AMS, UWILD, BP(30)

Bollard pull ..... 31tonnes  
 Main engine(s)

Make ..... Caterpillar  
 Model ..... C32  
 Number ..... 2  
 Output of each engine ..... 970kW

Propeller(s)

Material ..... NiAlBr  
 Manufacturer ..... Veth  
 Number ..... 2  
 Fixed/controllable pitch ..... Fixed  
 Diameter ..... 1,700mm  
 Speed ..... 352rpm  
 Open or nozzled ..... Nozzled

Alternator(s)

Make ..... Caterpillar C4.4  
 Number ..... 2  
 Output of each set ..... 86kWe

Deck machinery

1 x DTM TW-E winch, 100kN  
 1 x DTH towing hook, 30tonnes

Onboard capacities

Fuel oil ..... 33,500litres  
 Fresh water ..... 5,700litres  
 Ballast water ..... 10,200litres

Complement

Crew ..... 4  
 Passengers ..... 0  
 Number of cabins ..... 2